

High-Dimensional Granger Causality Tests with an Application to VIX and News*

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Abstract

We study Granger causality testing for high-dimensional time series using regularized regressions. To perform proper inference, we rely on heteroskedasticity and autocorrelation consistent (HAC) estimation of the asymptotic variance and develop the inferential theory in the high-dimensional setting. To recognize the time-series data structures, we focus on the sparse-group LASSO (sg-LASSO) estimator, which includes the LASSO and the group LASSO as special cases. We establish the debiased central limit theorem for low-dimensional groups of regression coefficients and study the HAC estimator of the long-run variance based on the sg-LASSO residuals. This leads to valid time-series inference for individual regression coefficients as well as groups, including Granger causality tests. The treatment relies on a new Fuk–Nagaev inequality for a class of τ -mixing processes with heavier than Gaussian tails, which is of independent interest. In an empirical application, we study the Granger causal relationship between the VIX and financial news.

Key words: fat tails, Fuk–Nagaev inequality, Granger causality, HAC estimator, high-dimensional time series, inference, sparse-group LASSO

JEL classification: C10, C12, C22

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Modern time-series analysis is increasingly using high-dimensional datasets, typically available at different frequencies. Conventional time series are often supplemented with non-traditional data, such as the high-dimensional data coming from the natural language processing. For instance, [Bybee et al. \(2020\)](#) extract 180 topic attention series from over 800,000 daily *Wall Street Journal* news articles during 1984–2017 that have been shown by [Babii, Ghysels, and Striaukas \(2022\)](#) to be a useful supplement to more traditional macro-economic and financial datasets for nowcasting U.S. GDP growth.

In his seminal paper, [Granger \(1969\)](#) defined causality in terms of high-dimensional time-series data. His formal definition, see [Granger \(1969\)](#), Definition 1, considered all the information accumulated in the universe up to time $t - 1$ (a process he called U_t) and examined predictability using U_t with and without a specific series of interest Y_t . It is still an open question how to implement Granger's test in a high-dimensional time-series setting. This article aims to do this via regularized regressions using heteroskedasticity and autocorrelation consistent (HAC)-based inference. In a sense, we are trying to implement Granger's original idea of causality.¹

It is worth relating our work to the existing literature on Granger causality with high-dimensional data. Various dimensionality reduction schemes have been considered. For example, [Box and Tiao \(1977\)](#) used canonical correlation analysis and [Peña and Box \(1987\)](#) and [Stock and Watson \(2002\)](#) proposed factor models and principle component analysis. [Koop \(2013\)](#) analyzed large dimensional Bayesian vector autoregressive (VAR) models. More closely related to our paper are [Yuan and Lin \(2006\)](#), [Simon et al. \(2013\)](#), [Skripnikov and Michailidis \(2019\)](#), [Nicholson et al. \(2020\)](#), and [Babii, Ghysels, and Striaukas \(2022\)](#), who look at structured sparsity approaches without doing inference. Granger causality with sparsity and inference also appeared in a number of papers. [Wilms, Gelper, and Croux \(2016\)](#) use bootstrap but ignore post-selection issues, while [Hecq, Margaritella, and Smeekes \(2019\)](#) extend the post-double selection approach of [Belloni, Chernozhukov, and Hansen \(2014\)](#) to Granger causality testing in linear sparse high-dimensional VAR. Finally, [Ghysels, Hill, and Motegi \(2020\)](#) propose a Granger causality test based on a seemingly overlooked, but simple, dimension reduction technique. The procedure involves multiple parsimonious regression models where key regressors are split across simple regressions. Each parsimonious regression model has one key regressor and other regressors not associated with the null hypothesis. The test is based on the maximum of the squared parameters of the key regressors.

Following [Babii, Ghysels, and Striaukas \(2022\)](#), we focus on the structured sparsity approach based on the sparse-group LASSO (sg-LASSO) regularization for the high-dimensional time-series analysis. The sg-LASSO allows capturing the group structures present in high-dimensional time-series regressions where a single covariate with its lags constitutes a group. Alternatively, we can also combine covariates of similar nature in groups. An attractive feature of this estimator is that it encompasses the LASSO and the group LASSO as special cases; hence, it allows improving upon the unstructured LASSO in the high-dimensional time-series setting. At the same time, the sg-LASSO can learn the distribution of time-series lags in a data-driven way solving elegantly the model selection

1 There exists an extensive literature on causal inference with machine-learning methods within the *static* Neyman–Rubin's potential outcomes framework; see [Athey and Imbens \(2019\)](#) for the excellent review and further references.

problem that dates back to Fisher (1937).² In particular, the group structure can also accommodate data sampled at different frequencies, as discussed in detail by Babii, Ghysels, and Striaukas (2022).

The proper inference for time-series data relies on the HAC estimation of the long-run variance; see Eicker (1963), Huber (1967), White (1980), Gallant (1987), Newey and West (1987), and Andrews (1991), among others.³ Despite the increasing popularity of the LASSO in finance and more generally in time-series empirical research, to the best of our knowledge, the validity of HAC-based inference for LASSO has not been established in the relevant literature.⁴ The HAC-based inference is robust to the model misspecification and leads to the valid Granger causality tests even when the fitted regression function has only projection interpretation, which is the case for the projection-based definition of the Granger causality considered in Granger (1969). Developing the asymptotic theory for the linear projection model with autoregressive lags and covariates, however, is challenging because the underlying processes are typically *not* β -mixing.⁵

In this article, we obtain the debiased central limit theorem with explicit bias correction for the sg-LASSO estimator and time-series data, which extends van de Geer et al. (2014) and to the best of our knowledge is new. Next, we establish the formal statistical properties of the HAC estimator based on the sg-LASSO residuals in the high-dimensional environment when the number of covariates can increase faster than the sample size. The convergence rate of the HAC estimator can be affected by the tails and the persistence of the data, which is a new phenomenon compared with low-dimensional regressions. For the practical implementation, this implies that the optimal choice of the bandwidth parameter for the HAC estimator should scale appropriately with the number of covariates, the tails, and the persistence of the data. These results allow us to perform inference for groups of coefficients, including the (mixed-frequency) Granger causality tests.

Our asymptotic theory applies to the heavy-tailed time-series data, which is often observed in financial and economic applications. To that end, we establish a new Fuk-Nagaev inequality, see Fuk and Nagaev (1971), for τ -mixing processes with *polynomial tails*. The class of τ -mixing processes is flexible enough for developing the asymptotic theory for the linear projection model and, at the same time, it contains the class of α -mixing processes as a special case.

- 2 The distributed lag literature can be traced back to Fisher (1925); see also Almon (1965), Sims (1971), and Shiller (1973), as well as the more recent MIDAS approach in Ghysels, Santa-Clara, and Valkanov (2006), Ghysels, Sinko, and Valkanov (2007), and Andreou, Ghysels, and Kourtellis (2013).
- 3 For stationary time series, the HAC estimation of the long-run variance is the same problem as the estimation of the value of the spectral density at zero, which itself has even longer history dating back to the smoothed periodogram estimators; see Daniell (1946), Bartlett (1948), and Parzen (1957).
- 4 See Krampe, Kreiss, and Paparoditis (2021) and Chernozhukov et al. (2021) for LASSO inference and causal Bernoulli shifts with independent innovations and Adamek, Smeekes, and Wilms (2020) for near-epoch dependent processes; see also Feng, Giglio, and Xiu (2020) for an asset pricing application; Belloni, Chernozhukov, and Hansen (2014) and van de Geer et al. (2014) for i.i.d. data; and Chiang and Sasaki (2019) for exchangeable arrays.
- 5 More generally, it is known that the linear transformations based on infinitely many lags do not preserve the α - or β -mixing property.

The paper is organized as follows. We start with the large sample approximation to the distribution of the sg-LASSO estimator (and as a consequence of the LASSO and the group LASSO) with τ -mixing data in Section 1. The reader interested in the practical implementation of the Granger causality test can refer directly to Section 1.6. Next, we consider the HAC estimator of the asymptotic long-run variance based on the sg-LASSO residuals and study the inference for groups of regression coefficients. In Section 2, we establish a suitable version of the Fuk–Nagaev inequality for τ -mixing processes. We report on a Monte Carlo study in Section 3 which provides further insights about the validity of our theoretical analysis in finite-sample settings typically encountered in empirical applications. Section 4 covers an empirical application examining the Granger causal relations between the VIX and financial news. Conclusions appear in Section 5. Proofs and [supplementary results](#) appear in the Appendix and the [Online Appendix](#).

Notation:

For a random variable $X \in \mathbf{R}$ and $q \geq 1$, let $\|X\|_q = (\mathbb{E}|X|^q)^{1/q}$ be its L_q norm. For a positive integer p , we put $[p] = \{1, 2, \dots, p\}$. For $G \subset [p]$ and $\Delta \in \mathbf{R}^p$, we also use $\Delta_G = \{\Delta_j : j \in G\}$ to denote the sub-vector of Δ indexed by G . Let $\mathcal{G} = \{G_g : g \geq 1\}$ be a partition of $[p]$ defining groups. For a vector of regression coefficients $\beta \in \mathbf{R}^p$, the sparse-group structure is described by a pair $(S_0, \mathcal{G}_0)$, where $S_0 = \{j \in [p] : \beta_j \neq 0\}$ is the support of β and $\mathcal{G}_0 = \{G \in \mathcal{G} : \beta_G \neq 0\}$ is its group support.

For $b \in \mathbf{R}^p$ and $q \geq 1$, the ℓ_q norm is denoted $|b|_q = (\sum_{j \in [p]} |b_j|^q)^{1/q}$ if $q < \infty$ and $|b|_\infty = \max_{j \in [p]} |b_j|$ if $q = \infty$. For $\mathbf{u}, \mathbf{v} \in \mathbf{R}^T$, the empirical inner product is defined as $\langle \mathbf{u}, \mathbf{v} \rangle_T = \frac{1}{T} \sum_{t=1}^T u_t v_t$ with the induced empirical norm $\|\cdot\|_T^2 = \langle \cdot, \cdot \rangle_T = \|\cdot\|_2^2 / T$. For a symmetric $p \times p$ matrix A , let $\text{vech}(A) \in \mathbf{R}^{p(p+1)/2}$ be its vectorization consisting of the lower triangular and the diagonal part. Let A_G be a sub-matrix consisting of rows of A corresponding to indices in $G \subset [p]$. If $G = \{j\}$ for some $j \in [p]$, then we simply put $A_G = A_j$. For $p \times p$ matrices A and B , let $\langle A, B \rangle = \text{trace}(A^\top B)$ be the inner product and let $\|A\| = \sqrt{\langle A, A \rangle}$ be the corresponding Frobenius norm. We will also use the following matrix norm $\|A\|_\infty = \max_{j \in [p]} |A_j|_1$. For $a, b \in \mathbf{R}$, we put $a \vee b = \max\{a, b\}$ and $a \wedge b = \min\{a, b\}$. Lastly, we write $a_n \lesssim b_n$ if there exists a (sufficiently large) absolute constant C such that $a_n \leq C b_n$ for all $n \geq 1$ and $a_n \sim b_n$ if $a_n \lesssim b_n$ and $b_n \lesssim a_n$.

1 HAC-Based Inference for sg-LASSO

This section covers the HAC-based inference for the sg-LASSO estimator. In the first subsection, we introduce the high-dimensional linear projection model and the sg-LASSO estimator. Next, we introduce the τ -mixing and other relevant assumptions. The following three subsections cover the debiased central limit theorem and the HAC estimator based on the sg-LASSO residuals. Lastly, we discuss the practical implementation of the high-dimensional Wald-type test for Granger causality.

1.1 High-Dimensional Time-Series Regressions and sg-LASSO

Consider a generic linear projection model

$$y_t = \sum_{j=1}^{\infty} \beta_j x_{t,j} + u_t, \quad \mathbb{E}[u_t x_{t,j}] = 0, \quad \forall j \geq 1, \quad t \in \mathbf{Z},$$

where $(y_t)_{t \in \mathbb{Z}}$ is a real-valued stochastic process and predictors $\{x_{t,j} : j \geq 1\}$ may include the intercept, time-varying covariates, (mixed-frequency) lags of covariates up to a certain order, as well as lags of the dependent variable. For a sample of size T , in the vector notation, we write

$$\mathbf{y} = \mathbf{m} + \mathbf{u},$$

where $\mathbf{y} = (y_1, \dots, y_T)^\top$, $\mathbf{m} = (m_1, \dots, m_T)^\top$ with $m_t = \sum_{j=1}^{\infty} \beta_j x_{t,j}$, and $\mathbf{u} = (u_1, \dots, u_T)^\top$. We approximate m_t with $x_t^\top \beta = \sum_{j=1}^p \beta_j x_{t,j}$ and put $\mathbf{X}\beta$, where $\mathbf{X}$ is $T \times p$ design matrix and $\beta \in \mathbb{R}^p$ is the unknown projection parameter. This approximation can be constructed from lagged values of y_t , some covariates, as well as lagged values of covariates measured at a higher frequency, in which case we obtain the autoregressive distributed lag mixed-frequency data sampling model (ARDL-MIDAS) described as

$$\phi(L)y_t = \sum_{k=1}^K \psi(L^{1/m}; \beta_k) x_{t,k} + u_t,$$

where $\phi(L) = I - \rho_1 L - \rho_2 L^2 - \dots - \rho_J L^J$ is a low-frequency lag polynomial and the MIDAS part $\psi(L^{1/m}; \beta_k) x_{t,k} = \frac{1}{m} \sum_{j=1}^m \beta_{k,j} x_{t-(j-1)/m,k}$ is a high-frequency lag polynomial; see [Andreou, Ghysels, and Kourtellis \(2013\)](#) and [Babii, Ghysels, and Striaukas \(2022\)](#). Note that when $m=1$, we have all data sampled at the same frequency and recover the standard ARDL model. The ARDL-MIDAS regression has a group structure where a single group is defined as all lags of $x_{t,k}$, or all lags of y_t and following [Babii, Ghysels, and Striaukas \(2022\)](#), we focus on the sg-LASSO regularized estimator.⁶ The leading example here is the MIDAS regression involving the projection of future low-frequency series onto its own lags and lags of high-frequency data aggregated via some dictionary, for example, the set of Legendre polynomials. The setup also covers what is sometimes called the reverse MIDAS, see [Forni, Guérin, and Marcellino \(2018\)](#), and mixed-frequency VAR, see [Ghysels \(2016\)](#), involving the projection of high-frequency data onto its own (high-frequency) lags and low-frequency data. Such regressions, which appear in the empirical application of the paper, simply amount to a different group structure.

The sg-LASSO, denoted $\hat{\beta}$, solves the regularized least-squares problem

$$\min_{b \in \mathbb{R}^p} \|\mathbf{y} - \mathbf{X}b\|_T^2 + 2\lambda \Omega(b), \quad (1)$$

with the penalty function

$$\Omega(b) = \alpha \|b\|_1 + (1 - \alpha) \|b\|_{2,1},$$

where $\|b\|_1 = \sum_{j=1}^p |b_j|$ is the ℓ_1 norm corresponding to the LASSO penalty, $\|b\|_{2,1} = \sum_{G \in \mathcal{G}} \|b_G\|_2$ is the group LASSO penalty, and the group structure $\mathcal{G}$ is a partition of $[p] = \{1, 2, \dots, p\}$ specified by the econometrician.

1.2 τ -Mixing and Other Assumptions

We measure the persistence of a series with τ -mixing coefficients. For a σ -algebra $\mathcal{M}$ and a random vector $\zeta \in \mathbb{R}^l$, put

6 The sg-LASSO estimator allows selecting groups and important group members at the same time.

$$\tau(\mathcal{M}, \xi) = \left\| \sup_{f \in \text{Lip}_1} |\mathbb{E}(f(\xi)|\mathcal{M}) - \mathbb{E}(f(\xi))| \right\|_1,$$

where $\text{Lip}_1 = \{f: \mathbf{R}^l \rightarrow \mathbf{R} : |f(x) - f(y)| \leq |x - y|_1\}$ is a set of 1-Lipschitz functions. Let $(\xi_t)_{t \in \mathbb{Z}}$ be a stochastic process and let $\mathcal{M}_t = \sigma(\xi_t, \xi_{t-1}, \dots)$ be its natural filtration. The τ -mixing coefficient is defined as

$$\tau_k = \sup_{j \geq 1} \frac{1}{j} \sup_{t+k \leq t_1 < \dots < t_j} \tau(\mathcal{M}_t, (\xi_{t_1}, \dots, \xi_{t_j})), \quad k \geq 0,$$

where the supremum is taken over t and $(t_1, \dots, t_j)$. The process is called τ -mixing if $\tau_k \downarrow 0$ as $k \uparrow \infty$; see Lemma A.1.1 for the comparison of this coefficient to the mixingale and the α -mixing coefficients as well as [Dedecker and Prieur \(2004, 2005\)](#) and [Dedecker et al. \(2007\)](#) for more details. The following assumptions impose tail and moment conditions.

Assumption 2.1 (Data): *The process $(u_t, x_t)_{t \in \mathbb{Z}}$ is stationary and such that (i) $\|u_t\|_q < \infty$ and $\max_{j \in [p]} \|x_{t,j}\|_r = O(1)$ for some $q > 2r/(r-2)$ and $r > 4$; (ii) for every $j, l \in [p]$, the τ -mixing coefficients of $(u_t x_{t,j})_{t \in \mathbb{Z}}$ and $(x_{t,j} x_{t,l})_{t \in \mathbb{Z}}$ are $\tau_k \leq ck^{-a}$ and $\bar{\tau}_k \leq ck^{-b}$ for all $k \geq 0$ and some universal constants $c > 0$, $a > (\varsigma - 1)/(\varsigma - 2)$, $b > (r - 2)/(r - 4)$, and $\varsigma = qr/(q + r)$.*

Assumption 2.1 can be relaxed to non-stationary data with stable variances of partial sums at the cost of heavier notation. It allows for heavy-tailed and persistent data. For instance, it requires that either both covariates and the error process have at least $4 + \epsilon$ finite moments, or that the error process has at least $2 + \epsilon$ finite moments, whenever covariates are sufficiently integrable. It is also known that the τ -mixing coefficients decline exponentially fast for geometrically ergodic Markov chains, including the stationary AR(1) process, so condition (ii) allows for relatively persistent data; see also [Babii, Ghysels, and Striaukus \(2022\)](#) for verification of these conditions for a heavy-tailed autoregressive model with covariates. Next, we require that the covariance matrix of covariates is invertible.

Assumption 2.2 (Covariance): *There exists a universal constant $\gamma > 0$ such that the smallest eigenvalue of $\Sigma = \mathbb{E}[x_t x_t^\top]$ is bounded away from zero by γ .*

Assumption 2.2 ensures that the precision matrix $\Theta = \Sigma^{-1}$ exists and rules out perfect multicollinearity. It also requires that the smallest eigenvalue of Σ is bounded away from zero by γ independently of the dimension p , which is the case, for example, for the spiked identity and the Toeplitz covariance structures. Strictly speaking this condition can be relaxed to $\gamma \downarrow 0$ as $p \uparrow \infty$ at the cost of slower convergence rates and more involved conditions on rates, in which case γ can be interpreted as a measure of ill-posedness; see [Carrasco, Florens, and Renault \(2007\)](#). The next assumption describes the rate of the regularization parameter, which is governed by the Fuk–Nagaev inequality; see Theorem 3.1 and [Equation \(7\)](#).

Assumption 2.3 (Regularization): *For some $\delta \in (0, 1)$*

$$\lambda \sim \left(\frac{p}{\delta T^{\kappa-1}} \right)^{1/\kappa} \vee \sqrt{\frac{\log(8p/\delta)}{T}},$$

where $\kappa = ((a + 1)\varsigma - 1)/(a + \varsigma - 1)$, where a, ς are as in Assumption 2.1.

In practice, we recommend selecting the tuning parameter in a data-driven way. It is beyond the scope of the present paper to study properties of estimators with data-driven tuning parameters; see [Chetverikov, Liao, and Chernozhukov \(2021\)](#) for this type of analysis with i.i.d. data. Lastly, we impose the following condition on the misspecification error, the number of covariates p , the sparsity constant $\sqrt{s_x} = \alpha\sqrt{|S_0|} + (1 - \alpha)\sqrt{|G_0|}$, and the sample size T .

Assumption 2.4: (i) $\|\mathbf{m} - \mathbf{X}\beta\|_T^2 = O_P(s_x\lambda^2)$; (ii) $s_x^\mu p^2 T^{1-\mu} \rightarrow 0$ and $p^2 \exp(-cT/s_x^2) \rightarrow 0$ as $T \rightarrow \infty$, where s_x is the effective sparsity of β and $\mu = ((b+1)r - 2)/(r + 2(b-1))$.

The effective sparsity constant $\sqrt{s_x}$ is a linear combination of the sparsity $|S_0|$ (number of non-zero coefficients) and the group sparsity $|G_0|$ (number of active groups). It reflects the finite-sample advantages of imposing the sparse-group structure, as $|G_0|$ can be significantly smaller than $|S_0|$ that appears in the theory of the standard LASSO estimator. Throughout the paper, we assume that the groups have fixed size, which is well justified in time-series applications of interest.

The four assumptions listed above are needed for the prediction and estimation consistency of the sg-LASSO estimator; see [Theorem OA.1 in the Online Appendix](#).

1.3 Nodewise LASSO

The LASSO estimator $\hat{\beta}$ is biased and may have a complicated distribution. Nonetheless, the distribution of subvectors of debiased LASSO estimator $\hat{\beta} + B$ of fixed size can be approximated with the Gaussian distribution. The bias-correction term $B = \hat{\Theta}\mathbf{X}^\top(\mathbf{y} - \mathbf{X}\hat{\beta})/T$ depends on the estimator of the precision matrix $\Theta = \Sigma^{-1}$, where $\Sigma = \mathbb{E}[\mathbf{X}\mathbf{X}^\top]$ is a covariance matrix of $\mathbf{X} \in \mathbf{R}^p$. In this section, we discuss the nodewise LASSO estimator of Θ following [Meinshausen and Bühlmann \(2006\)](#) and [van de Geer et al. \(2014\)](#). The estimator is based on the observation that the covariance matrix of the partitioned vector $\mathbf{X} = (X_j, \mathbf{X}_{-j}^\top)^\top \in \mathbf{R} \times \mathbf{R}^{p-1}$ can be written as

$$\Sigma = \mathbb{E}[\mathbf{X}\mathbf{X}^\top] = \begin{pmatrix} \Sigma_{jj} & \Sigma_{j,-j} \\ \Sigma_{-j,j} & \Sigma_{-j,-j} \end{pmatrix},$$

where $\Sigma_{jj} = \mathbb{E}[X_j^2]$ and all other elements similarly defined. By the partitioned inverse formula, the first row of the precision matrix $\Theta = \Sigma^{-1}$ is

$$\Theta_j = \sigma_j^{-2} \begin{pmatrix} 1 & -\gamma_j^\top \end{pmatrix},$$

where $\gamma_j = \Sigma_{-j,-j}^{-1}\Sigma_{-j,j}$ is the projection coefficient in the regression of X_j on $\mathbf{X}_{-j}$

$$X_j = \mathbf{X}_{-j}^\top \gamma_j + v_j, \quad \mathbb{E}[X_{-j}v_j] = 0, \quad (2)$$

and $\sigma_j^2 = \Sigma_{jj} - \Sigma_{j,-j}\gamma_j = \mathbb{E}[v_j^2]$ is the variance of the projection error.⁷ This suggests estimating the first row of the precision matrix as $\hat{\Theta}_j = \hat{\sigma}_j^{-2} \begin{pmatrix} 1 & -\hat{\gamma}_j^\top \end{pmatrix}$ with $\hat{\gamma}_j$ solving

⁷ To ensure that the projection coefficient is well defined and does not change with the dimension of the model p , we can consider the limiting linear projection model and take into account the approximation error.

$$\min_{\gamma \in \mathbb{R}^{p-1}} \|\mathbf{X}_j - \mathbf{X}_{-j}\gamma\|_T^2 + 2\lambda_j|\gamma|_1$$

and

$$\hat{\sigma}_j^2 = \|\mathbf{X}_j - \mathbf{X}_{-j}\hat{\gamma}_j\|_T^2 + \lambda_j|\hat{\gamma}_j|,$$

where $\mathbf{X}_j \in \mathbb{R}^T$ is the column vector of observations of $x_j \in \mathbb{R}$ and $\mathbf{X}_{-j}$ is the $T \times (p-1)$ matrix of observations of $x_{-j} \in \mathbb{R}^{p-1}$. In the matrix notation, the nodewise LASSO estimator of Θ can be written then as $\hat{\Theta} = \hat{B}^{-1}\hat{C}$ with

$$\hat{C} = \begin{pmatrix} 1 & -\hat{\gamma}_{1,1} & \cdots & -\hat{\gamma}_{1,p-1} \\ -\hat{\gamma}_{2,1} & 1 & \cdots & -\hat{\gamma}_{2,p-1} \\ \vdots & \vdots & \ddots & \vdots \\ -\hat{\gamma}_{p-1,1} & \cdots & -\hat{\gamma}_{p-1,p-1} & 1 \end{pmatrix} \quad \text{and} \quad \hat{B} = \text{diag}(\hat{\sigma}_1^2, \dots, \hat{\sigma}_p^2).$$

Let $v_{t,j}$ be the projection error in j th nodewise LASSO regression. Put also $s = s_x \vee S$, $S = \max_{j \in G} S_j$, where S_j is the number of non-zero coefficients in the j th row of Θ . The following assumption describes an additional set of sufficient conditions for the debiased central limit theorem.

Assumption 2.5: (i) $\sup_x \mathbb{E}[u_t^2 | x_t = x] = O(1)$; (ii) $\|\Theta_G\|_\infty = O(1)$ for some $G \subset [p]$ of fixed size; (iii) the long-run variances of $(u_t^2)_{t \in \mathbb{Z}}$ and $(v_{t,j}^2)_{t \in \mathbb{Z}}$ exist for every $j \in G$; (iv) $s^2 \log^2 p / T \rightarrow 0$ and $p / \sqrt{T^{k-2} \log^k p} \rightarrow 0$; (v) $\|\mathbf{m} - \mathbf{X}\beta\|_T = o_P(T^{-1/2})$; (vi) for every $j, l \in [p]$ and $k \geq 0$, the τ -mixing coefficients of $(u_t u_{t+k} x_{t,j} x_{t+k,l})_{t \in \mathbb{Z}}$ are $\tilde{\tau}_t \leq ct^{-d}$ for some universal constants $c > 0$ and $d > 1$.

Assumption (i) requires that the conditional variance of the regression error is bounded. Condition (ii) requires that the rows of the precision matrix have bounded ℓ_1 norm and is a plausible assumption in the high-dimensional setting, where the inverse covariance matrix is often sparse, for example, in the Gaussian graphical model. Condition (iii) is a mild restriction needed for the consistency of the sample variance of regression errors. The rate imposed on the sparsity constant, $s^2 \log^2 p / T \rightarrow 0$, is also used in van de Geer et al. (2014), who assume that the regression errors are Gaussian; see their Corollary 2.1. On the other hand, the rate condition on the dimension $p / \sqrt{T^{k-2} \log^k p} \rightarrow 0$ is an additional condition needed in our setting when regression errors are not Gaussian and may only have a certain number of finite moments. Lastly, condition (v) is trivially satisfied when the projection coefficients are sparse and, more generally, it requires that the misspecification error vanishes asymptotically sufficiently fast. Conditions of this type are standard in the nonparametric literature.

It is also worth mentioning that Assumption 2.5 does not require that β or Θ are sparse, that is, we can have $s = O(p)$, in which case a larger T may be required. It is also known that one can use the alternative weak sparsity conditions defined as a small value of the ℓ_q norm of β with $q \in (0, 1)$; see van de Geer (2016), Section 2.10. Lastly, for simplicity of presentation, we state all conditions in terms of the largest sparsity constant $s = s_x \vee \max_{j \in G} S_j$. One can easily relax this at the cost of heavier notation.

1.4 Debiased CLT

Let $B = \hat{\Theta} \mathbf{X}^\top (\mathbf{y} - \mathbf{X}\hat{\beta}) / T$ denote the bias-correction for the sg-LASSO estimator, where $\hat{\Theta}$ is the nodewise LASSO estimator of the precision matrix Θ ; see the following subsection for more

details. The following result describes a large-sample approximation to the distribution of the debiased sg-LASSO estimator with serially correlated non-Gaussian regression errors.

Theorem 2.1: *Suppose that Assumptions 2.1–2.5 are satisfied for the sg-LASSO regression and for each nodewise LASSO regression $j \in G$. Then*

$$\sqrt{T}(\hat{\beta}_G + B_G - \beta_G) \xrightarrow{d} N(0, \Xi_G),$$

with the long-run variance $\Xi_G = \lim_{T \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \Theta_G X_t \right)$.

It is worth mentioning that since the group G has a fixed size and the rows of Θ have a finite ℓ_1 norm, the long-run variance Ξ_G exists under the maintained assumptions; see Proposition A.1.1 in the Appendix for a precise statement.

Theorem 2.1 extends [van de Geer et al. \(2014\)](#) to non-Gaussian, heavy-tailed, and persistent time-series data and describes the long-run asymptotic variance for the low-dimensional group of regression coefficients estimated with the sg-LASSO. One could also consider Gaussian approximations for groups of increasing size, which requires an appropriate high-dimensional Gaussian approximation result for τ -mixing processes and is left for future research; see [Chernozhukov, Chetverikov, and Kato \(2013\)](#) for a comprehensive review of related coupling results in the i.i.d. case.

Remark 2.1: *It is worth mentioning that the debiasing with explicit bias correction addresses the post-model selection issues, see [Leeb and Pötscher \(2005\)](#), and it is fairly straightforward to show that the convergence in Theorem 2.1 holds uniformly over the set of sparse vectors; see also [van de Geer et al. \(2014\)](#), Corollary 2.1, and the remark following that corollary.*

1.5 HAC Estimator

Next, we focus on the HAC estimator based on sg-LASSO residuals, covering the LASSO and the group LASSO as special cases. For a group $G \subset [p]$ of a fixed size, the HAC estimator of the long-run variance is

$$\hat{\Xi}_G \triangleq \sum_{|k| < T} K \left(\frac{k}{M_T} \right) \hat{\Gamma}_k, \quad (3)$$

where $\hat{\Gamma}_k = \hat{\Theta}_G \left(\frac{1}{T} \sum_{t=1}^{T-k} \hat{u}_t \hat{u}_{t+k} x_t x_{t+k}^\top \right) \hat{\Theta}_G^\top$, $\hat{u}_t$ is the sg-LASSO residual and $\hat{\Gamma}_{-k} = \hat{\Gamma}_k^\top$. The kernel function $K: \mathbf{R} \rightarrow [-1, 1]$ with $K(0) = 1$ puts less weight on more distant noisy covariances, while $M_T \uparrow \infty$ is a bandwidth (or lag truncation) parameter; see [Parzen \(1957\)](#), [Newey and West \(1987\)](#), and [Andrews \(1991\)](#). Several choices of the kernel function are possible; for example, the Parzen kernel is

$$K_{PR}(x) = \begin{cases} 1 - 6x^2 + 6|x|^3 & \text{for } 0 \leq |x| \leq 1/2, \\ 2(1 - |x|)^3 & \text{for } 1/2 \leq |x| \leq 1, \\ 0 & \text{otherwise.} \end{cases}$$

It is worth recalling that the Parzen and the Quadratic spectral kernels are high-order kernels that are superior to the Bartlett kernel, cf. Newey and West (1987); see the Appendix for more details on the choice of the kernel.

Note that under stationarity, the long-run variance in Theorem 2.1 simplifies to

$$\Xi_G \triangleq \sum_{k \in \mathbb{Z}} \Gamma_k,$$

where $\Gamma_k = \Theta_G \mathbb{E}[u_t x_t u_{t+k} x_{t+k}^\top] \Theta_G^\top$ and $\Gamma_{-k} = \Gamma_k^\top$. The following result characterizes the convergence rate of the HAC estimator pertaining to a group of regression coefficients $G \subset [p]$ based on the sg-LASSO residuals.

Theorem 2.2: *Suppose that Assumptions 2.1–2.5 are satisfied for the sg-LASSO regression and for each nodewise LASSO regression $j \in G$. Suppose also that Assumptions A.1.1 in the Appendix and OA.1.1 in the Online Appendix are satisfied for $V_t = (u_t v_{t,j} / \sigma_j^2)_{j \in G}$, $\kappa \geq \tilde{q}$ and that $s^\kappa p T^{1-4\kappa/5} \rightarrow 0$ as $M_T \rightarrow \infty$ and $T \rightarrow \infty$. Then*

$$\|\hat{\Xi}_G - \Xi_G\| = O_P \left(M_T \left(\frac{s p^{1/\kappa}}{T^{1-1/\kappa}} \vee s \sqrt{\frac{\log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} \right) + M_T^\zeta + T^{-(\zeta \wedge 1)} \right).$$

The first term in the inner parentheses is of the same order as the estimation error of the maximum between the estimation errors of the sg-LASSO and the nodewise LASSO. Theorem 2.2 suggests that the optimal choice of the bandwidth parameter should scale appropriately with the number of covariates p , the sparsity constant s , and the dependence-tails exponent κ . This contrasts sharply with the HAC theory for regressions without regularization developed in Andrews (1991), see also Li and Liao (2020), and allows for faster convergence rates of the HAC estimator. We deduce from Theorem 2.2 that the HAC estimator is consistent whenever $M_T \left(\frac{s p^{1/\kappa}}{T^{1-1/\kappa}} \vee s \sqrt{\frac{\log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} \right) \rightarrow 0$ as $M_T \rightarrow \infty$ and $T \rightarrow \infty$.

1.6 High-Dimensional Granger Causality Tests

We would like to test whether a series $(w_t)_{t \in \mathbb{Z}}$ Granger causes another series $(y_t)_{t \in \mathbb{Z}}$ at some horizon h . Consider the following projection equation:

$$y_{t+h} = c + \sum_{j \geq 1} z_{t,j} \gamma_j + \mathbf{w}_{t-1}^\top \alpha + u_t, \quad (4)$$

where $(z_{t,j})_{j \geq 1}$ is a high-dimensional set of controls representing “all the information in the universe available at time t ”; see Granger (1969), Section 4; $\alpha = (\alpha_1, \dots, \alpha_K)^\top$ and $\mathbf{w}_{t-1} = (w_{t-1}, \dots, w_{t-K})^\top$ are lags of $(w_t)_{t \in \mathbb{Z}}$. This high-dimensional set of controls may contain low- and/or high-frequency lags of $(y_t)_{t \in \mathbb{Z}}$ or some other series.

The null hypothesis that $(w_t)_{t \in \mathbb{Z}}$ does not Granger cause $(y_t)_{t \in \mathbb{Z}}$ and the corresponding alternative hypothesis are

$$H_0 : \alpha = 0 \quad \text{versus} \quad H_1 : \alpha \neq 0.$$

It is worth mentioning that our framework is based on the weakest notion of the Granger causality, as in Granger (1969) corresponding to the marginal improvement in time-series projections due to the information contained in w_t . A stronger notion of

Granger non-causality appears when projections are replaced by conditional means, so that the conditional mean y_t given w_t and all other available information does not depend on w_t . Yet, even stronger version of Granger non-causality pertains to the full conditional independence; see [Florens and Mouchart \(1982\)](#).

The Granger causality test is nested by our more general HAC-based inference framework. To that end, consider

$$y_{t+b} = \sum_{j \in G} \beta_j x_{t,j} + \sum_{j \in G^c} \beta_j x_{t,j} + u_t,$$

where we define $x_t^\top \beta_G = w_{t-1}^\top \alpha$ and $\sum_{j \in G^c} \beta_j x_{t,j} = c + \sum_{j \geq 1} z_{t,j} \gamma_j$, and $G \subset [p]$ is a relevant group of projection coefficients.

Therefore, testing the Granger causality amounts to

$$H_0 : \beta_G = 0 \quad \text{versus} \quad H_1 : \beta_G \neq 0.$$

Theorems 2.1 and 2.2 imply that, under H_0 , the asymptotic distribution of the Wald statistics is as follows:

$$W_T \triangleq T[(\hat{\beta}_G + B_G - \beta_G)]^\top \hat{\Xi}_G^{-1} [(\hat{\beta}_G + B_G - \beta_G)] \xrightarrow{d} \chi_{|G|}^2, \quad (5)$$

provided that consistency conditions for HAC estimator are satisfied. Note that $\hat{\Xi}_G$ is a positive definite matrix for commonly used kernel functions; see [Newey and West \(1987\)](#) and [Andrews \(1991\)](#). The Wald test rejects when $W_T > q_{1-\alpha}$, where $q_{1-\alpha}$ is the quantile of order $1 - \alpha$ of $\chi_{|G|}^2$.

The practical implementation of the Granger causality test is as follows:

1. Estimate $\alpha \in \mathbf{R}^K$ in [Equation \(4\)](#) using the sg-LASSO.
2. Compute the HAC estimator using the sg-LASSO residuals as in [Equation \(2\)](#).
3. Compute the bias-corrected Wald statistics for $H_0 : \alpha = 0$ in [Equation \(5\)](#). The bias correction requires fitting K nodewise LASSO regressions described in Section 1.3.
4. Reject the Granger non-causality if $W_T > q_{1-\alpha}$, where $q_{1-\alpha}$ is the $1 - \alpha$ quantile of χ_K^2 and do not reject otherwise.

To tune the sg-LASSO estimator and the nodewise LASSO regressions, we consider the following time-series cross-validation. For some $l \in \mathbf{N}$ and each $t = 1, \dots, T$

1. If $t > l + 1$ and $t < T - l$, use observations $I_{t,l} = \{1, \dots, t - l - 1, t + l + 1, \dots, T\}$ to estimate the sg-LASSO parameter, denoted $\hat{\beta}_{-t,l}(\lambda, \alpha)$. For $t = 1, \dots, l + 1$, use $I_{t,l} = \{t + l + 1, \dots, T\}$ as the training sample. Similarly, for $t = T - l, \dots, T$, use $I_{t,l} = \{1, \dots, T - l - 1\}$ as the training sample.
2. Use the left-out observations to test the model

$$CV(\lambda, \alpha) = \frac{1}{T} \sum_{t=1}^T (Y_t - X_t^\top \hat{\beta}_{-t,l}(\lambda, \alpha))^2$$

3. Minimize $CV(\lambda, \alpha)$ with respect to (λ, α) .

Note that for $l=0$, the procedure coincides with the usual leave-one-out cross-validation. For $l \geq 1$, there is a gap of l observations separating the test and the training samples, which aims to reduce the dependence between the two. To reduce the computational burden, in the spirit of the K -fold cross-validation, we can draw randomly a sub-sample $I \subset [T]$ of size K and minimize

$$CV_K(\lambda, \alpha) = \frac{1}{K} \sum_{t \in I} (Y_t - X_t^\top \hat{\beta}_{-t}(\lambda, \alpha))^2$$

instead. Figure 1 illustrates this procedure with gap $l=3$ observations. We note that the training observations (red) are separated from a single test observation (blue) by a gap of three left-out observations (green) on each side. It is worth mentioning that similar procedures have been used to select the bandwidth parameter for nonparametric regression estimators with dependent data; see Chu and Marron (1991) and references therein. A variation of the “sample splitting with a gap” procedure for Tikhonov regularization is also considered in Babii (2021).

Lastly, for the HAC estimator, we propose the following “rule of thumb” to select the bandwidth parameter

$$M_T = \begin{cases} 1.3 \left(\frac{T}{\log p} \right)^{\frac{1}{1+\varsigma}}, & \text{sub-Gaussian data} \\ 1.3 \left(\frac{T^{2-2/q}}{p^{2/q}} \right)^{\frac{1}{1+\varsigma}}, & \text{heavy-tailed data,} \end{cases} \quad (6)$$

where ς is the order of the kernel ($\varsigma = 1$ for the Bartlett kernel and $\varsigma = 2$ for the Quadratic spectral and Parzen kernels), $q > 2$ is the number of finite moments in the data, T is the sample size, and p is the number of regressors. Note that for the Bartlett kernel in the sub-Gaussian case, we have $M_T = 1.3(T/\log p)^{1/2}$, which is similar to the “rule of thumb” $M_T = 1.3T^{1/2}$ proposed in Lazarus et al. (2018). We also recommend using the quadratic spectral kernel which has certain optimality properties for inference; see Sun and Yang (2020).

2 Fuk–Nagaev Inequality

In this section, we describe a suitable version of the Fuk–Nagaev concentration inequality for the maximum of high-dimensional sums. The inequality allows for the data with polynomial tails and τ -mixing coefficients decreasing at a polynomial rate. The following result does not require that the series is stationary.

Theorem 3.1: *Let $(\xi_t)_{t \in \mathbb{Z}}$ be a centered stochastic process in $\mathbb{R}^p$ such that (i) for some $q > 2$, $\max_{j \in [p], t \in [T]} \|\xi_{t,j}\|_q = O(1)$; (ii) for every $j \in [p]$, τ -mixing coefficients of $\xi_{t,j}$ satisfy $\tau_k^{(j)} \leq ck^{-a}$ for some universal constants $a, c > 0$. Then there exist $c_1, c_2 > 0$ such that for every $u > 0$*

$$\Pr \left(\left| \sum_{t=1}^T \xi_t \right|_{\infty} > u \right) \leq c_1 p T u^{-\kappa} + 4p \exp \left(-\frac{c_2 u^2}{B_T^2} \right),$$

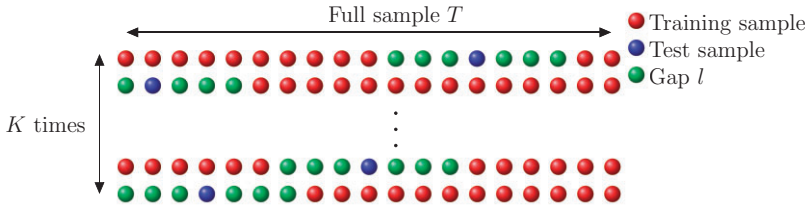


Figure 1. Time-series cross-validation scheme with $l=3$.

where⁸ $\kappa = ((a+1)q-1)/(a+q-1)$, $B_T^2 = \max_{j \in [p]} \sum_{t=1}^T \sum_{k=1}^T |\text{Cov}(\xi_{t,j}, \xi_{k,j})|$.

The inequality describes the mixture of the polynomial and Gaussian tails for the maximum of high-dimensional sums. In the limiting case of i.i.d. data, as $a \rightarrow \infty$, the dependence-tails exponent $\kappa \rightarrow q$, and we recover the inequality for the independent data stated in Fuk and Nagaev (1971), Corollary 4 for $p=1$. In this sense, the inequality in Theorem 3.1 is sharp. It is well-known that the Fuk–Nagaev inequality delivers sharper estimates of tail probabilities in contrast to Markov’s bound in conjunction with Rosenthal’s moment inequality (see Nagaev 1998). The proof relies on the blocking technique, see Bosq (1993), and the coupling inequality for τ -mixing sequences, see Dedecker and Prieur (2004), Lemma 5. In contrast to previous results, for example, Dedecker and Prieur (2004), Theorem 2, the inequality reflects the mixture of the polynomial and exponential tails.

For stationary processes, by Lemma A.1.2 in the Appendix, $B_T^2 = O(T)$ as long as $a > (q-1)/(q-2)$, whence we obtain from Theorem 3.1 that for every $\delta \in (0, 1)$

$$\Pr \left(\left| \frac{1}{T} \sum_{t=1}^T \xi_t \right|_{\infty} \leq C \left(\frac{p}{\delta T^{\kappa-1}} \right)^{1/\kappa} \vee \sqrt{\frac{\log(8p/\delta)}{T}} \right) \geq 1 - \delta, \quad (7)$$

where $C > 0$ is some finite universal constant.

3 Monte Carlo Experiments

In this section, we aim to assess the debiased HAC-based inferences for the low-dimensional parameter in a high-dimensional data setting. To that end, we report results for three different scenarios.

For the first scenario (denoted DGP 1), we draw independent AR(1) covariates $\{x_{t,j}, j \in [p]\}$

$$x_{t,j} = \rho x_{t-1,j} + \epsilon_{t,j},$$

and the regression error follows the AR(1) process

$$u_t = \rho u_{t-1} + \nu_t,$$

8 It is worth mentioning that the notation in this section is specific to generic stochastic processes and is independent from the rest of the paper. B_T here denotes the variance of partial sums and not the bias correction term of the LASSO estimator.

where the errors $\epsilon_{t,j} (j \in [p])$, $\nu_t \sim_{\text{i.i.d.}} N(0, 1)$. We set the persistence parameter $\rho = 0.6$. For the second heavy-tailed data scenario (DGP 2) the errors $\epsilon_t \sim_{\text{i.i.d.}} \text{Student-}t(5)$. Lastly, in the third scenario (DGP 3), the covariates follow a multivariate VAR process

$$X_t = \Phi X_{t-1} + \epsilon_t.$$

The sample size is $T \in \{100, 500\}$ and the number of covariates is $p \in \{10, 200\}$. For DGP 1 and DGP 2, we apply the LASSO estimator to estimate coefficients $\hat{\beta}$. For DGP 3, we use the block structure of the VAR to group covariates. Throughout the experiment, we choose all tuning parameters using the time-series cross-validation described in Section 1.6 with $K = 20$ and $l = 5$.

We report the empirical rejection frequency (ERF) for the Granger causality test based on the HAC estimator with two different kernel functions, namely Parzen and quadratic spectral kernels. We set the bandwidth parameter according to the rules appearing in Equation (6). For DGP 1 and DGP 3, we apply the sub-Gaussian rule, while for DGP 2 we use the heavy-tailed data rule with $q = 5$.

The vector of population regression coefficients β has the first five non-zero entries and all remaining entries are zero. To assess the performance, we scale the coefficient vector by multiplying it with a constant $c \in \{0, 1/2, 1\}$, that is, the coefficient for the relevant covariates is $c \times (1, 1, 1, 1, 1)$. For $c = 0$, the ERF shows the empirical size of the test for the nominal level of 5%, while $c \in \{1/2, 1\}$ the ERF shows the empirical power of the Granger causality test. For the larger scaling constant c , the alternatives are further away from the null hypothesis and therefore the Granger causality test is expected to perform better.

We estimate the long-run covariance matrix $\hat{\Xi}$ using the LASSO or sg-LASSO residuals, denoted $\hat{u}_t$. We use nodewise LASSO regressions to estimate the precision matrix Θ for which we also use time-series cross-validation to compute LASSO tuning parameter. The first step is to compute the scores $\hat{V}_t = \hat{u}_t x_t$, where $\hat{u}_t = y_t - x_t^\top \hat{\beta}$ and $\hat{\beta}$ is the LASSO estimator. Next, we compute the high-dimensional HAC estimator using the formula in Equation (3), applying the relevant bandwidth parameter based on the DGP. We compute the pivotal statistics for each MC experiment $i \in [N]$ and each coefficient $j \in [p]$ as $\text{pivot}_j^{(i)} \triangleq (\hat{\beta}_j^{(i)} + B_j^{(i)} - \beta_j) / \sqrt{\hat{\Xi}_{jj}^{(i)} / T}$, where $B_j^{(i)} = \hat{\Theta}_j^{(i)} \mathbf{X}^{\top(i)} \hat{\mathbf{u}}^{(i)} / T$ and $\hat{\mathbf{u}}^{(i)} = \mathbf{y}^{(i)} - \mathbf{X}^{(i)} \hat{\beta}^{(i)}$. Finally, we compute the ERF rates by checking whether or not a variable is significant. The number of Monte Carlo experiments is set to $N = 2000$.

We report the Monte Carlo simulation average ERF rates for each scaling constant $c \in \{0, 1/2, 1\}$ over the active set of regressors. Table 1 shows the results for sample sizes $T = 100$ and $T = 500$. The table has three panels, one for each DGP. We find that the empirical size approaches the nominal 5% level for all DGPs as the sample size increases. As expected, the empirical power of the Granger causality test also increases with the sample size. The best performance is for DGP 1, reported in Panel (a) of Table 1, whereas the performance deteriorates the most for the heavy-tailed DGP 2, as shown in Panel (b). Finally, the performance of the Granger causality tests for the DGP 3 looks like that of DGP 1. Hence, heavy-tailed data have more of an adverse effect on Granger causality testing than the multivariate structure of covariates. Interestingly, the performance deteriorates with the

Table 1. HAC-based inference simulation results

c, T, p		Parzen		Quadratic spectral	
		10	200	10	200
Panel (a)—DGP 1					
$c = 0$	$T = 100$	0.083	0.097	0.075	0.080
	$T = 500$	0.060	0.077	0.058	0.069
$c = 0.5$	$T = 100$	0.765	0.701	0.825	0.719
	$T = 500$	0.904	0.889	0.915	0.863
$c = 1$	$T = 100$	1.0000	0.953	1.0000	0.944
	$T = 500$	1.0000	1.0000	1.0000	1.0000
Panel (b)—DGP 2					
$c = 0$	$T = 100$	0.097	0.117	0.087	0.104
	$T = 500$	0.065	0.077	0.066	0.075
$c = 0.5$	$T = 100$	0.711	0.645	0.771	0.680
	$T = 500$	0.882	0.844	0.895	0.848
$c = 1$	$T = 100$	0.999	0.932	0.999	0.932
	$T = 500$	1.000	1.000	1.000	1.000
Panel (c)—DGP 3					
$c = 0$	$T = 100$	0.093	0.115	0.082	0.091
	$T = 500$	0.064	0.080	0.058	0.071
$c = 0.5$	$T = 100$	0.711	0.689	0.691	0.666
	$T = 500$	0.893	0.869	0.908	0.856
$c = 1$	$T = 100$	1.000	0.910	1.000	0.897
	$T = 500$	1.000	1.000	1.000	1.000

Notes: We report results for two kernel functions (Parzen and quadratic spectral (QS)) with the bandwidth parameters selected using the rules-of-thumb appearing in Equation (6). In addition, we report results for three data generating process: (a) Gaussian data, (b) Student- $t(5)$ data, and (c) Gaussian data with VAR model for covariates.

number of noisy regressors across all scenarios. The largest decrease in this case is for the DGP 2. Overall though, the simulation results corroborate our theoretical results.

4 Testing Granger Causality for VIX and News

The CBOE Volatility Index, known as the VIX, is a popular measure of market-based expectations of future volatility and is often referred to as the “fear index.” The VIX index quotes the expected annualized change in the S&P 500 index over the following 30 days, computed using a combination of derivatives pricing theory and options-market data.

There is a large literature studying the theoretical and empirical properties of the VIX and it is impossible to cite only a few papers to do justice to all the outstanding research

output on the topic. Focusing only on Granger causal patterns, several studies pertain to causality between the VIX and VIX futures. For example, [Bollen, O'Neill, and Whaley \(2017\)](#) suggest that the VIX futures lagged the VIX in the first few years after its introduction and show an increasing dominance of VIX futures over time. Along similar lines, [Shu and Zhang \(2012\)](#) study price discovery between the VIX futures and the spot VIX index finding evidence of a bi-directional causal pattern.

We study the causal relationship between financial news and the VIX. There is also a substantial literature on the impact of news releases on financial markets (e.g., [Anderson et al. 2003](#)). Traditionally, such analysis studies the behavior of asset prices pre- and post-release. News is usually quantified numerically via the surprise component measured as the difference between an expectation prior to the release and the announcement. In the age of machine learning, the characterization of news has been expanded into the textual analysis of news coverage. To paraphrase the title of [Gentzkow, Kelly, and Taddy \(2019\)](#), the text is treated as data. It is in this spirit that we conduct our high-dimensional Granger causality analysis between the VIX and news.

We use a data set from [Bybee et al. \(2020\)](#) which contains 180 news attention monthly series, all of which potentially Granger cause future U.S. equity market volatility.⁹ A table with the full list of news topics series appears in the [Online Appendix Table OA.1](#). We estimate the following time-series regression model

$$y_{t+1} = \psi(L^{1/m}; \beta)y_t + \sum_{k=1}^K \rho_k x_{t,k} + u_t, \quad t \in [T],$$

where y_{t+1} is the value of the VIX at the end of month $t + 1$, $\psi(L^{1/m}; \beta)y_t$ is a MIDAS polynomial of 22 daily VIX lags where the first lag is the last day of the month t , and $x_{t,k}$ is the k th news attention series. Note that we take only one lag for the news attention series to simplify the model (which is also an empirically justified simplification). The MIDAS polynomial of daily lags of the VIX involves Legendre polynomials of degree 3. Note that the specification is what is sometimes called a reverse MIDAS regression as mentioned earlier in the paper. Prior to estimating the regression model, we time demean the response and covariates such that the intercept is zero. We further standardize all covariates and the daily VIX lags to have a unit standard deviation.

We apply the sg-LASSO estimator to estimate the slope coefficients and nodewise LASSO regressions to estimate the precision matrix. To fully exploit the group sparsity of sg-LASSO, we group all high-frequency lags of daily VIX, see [Babii, Ghysels, and Striaukas \(2022\)](#), for further details on such grouping. The news attention series are monthly and we are interested in whether the most recent news Granger causes the VIX; hence, we don't apply the group structure along the time dimension. Instead, we group news attention series based on a broader theme available for each series; see [Bybee et al. \(2020\)](#) for further details. Namely, the data set contains 24 broader topics which group each of the 180 news attention series. As a robustness check, we also compute results including 12 lags of each news series. In this case, we apply Legendre polynomials of degree 3 and group all lags pertaining to each news series.

9 We downloaded daily VIX data from St. Louis Fed FRED database and took the end-of-month values. The FRED mnemonic for the VIX is VIXCLS.

4.1 Main Results

We report the p -values for series that appear to be significant at least at the 5% significance level for a range of bandwidth values and for two kernel functions, namely Parzen and Quadratic Spectral. For the bandwidth parameter, we follow the rules-of-thumb appearing in Equation (6), with M_T^{sG} for sub-Gaussian data and M_T^{bt} for heavy-tailed data with $q = 5$. The sample starts in January 1990 and ends in June 2017, determined by the availability of the textual analysis data. We also report results when we truncate the sample at December 2007. The sg-LASSO tuning parameters are selected via the time-series cross-validation; see Section 3. Similarly, we tune the nodewise LASSO regressions for the precision matrix estimation using the same method. The results appear in Table 2. The table contains three panels with results for the structured sg-LASSO estimator using the full data sample in Panel (a), the sample prior to the global financial crisis in Panel (b), and a different lag structure for the news series in Panel (c).

4.1.1 Granger causality of news topics

The lagged daily VIX and the *Financial crisis* news series are highly significant at the 1% significance level according to the results in Table 2 Panel (a) for the full sample and Panel (b) for the pre-Great Depression sample, irrespective of the choice of kernel function. In the model with 12 lags for each news series, the covariate *Financial crisis* remains significant at the 5% significance level (Table 2 Panel (c)). The *Recession* news series appears to be significant in Panel (a), but it is not significant if we exclude the data starting from January 2008 onward (see Panel (b)) or if we use the model with 12 lags for each news series reported in Panel (c). Therefore, the significance of the *Recession* news series seems to be affected by the sample choice or model specification, while the *Financial crisis* news series remains significant irrespective of these choices.

4.1.2 Bi-directional Granger causality

We also test whether the daily VIX Granger causes the *Financial crisis* news series. For this we run the following MIDAS regression model

$$x_{t+1,j} = \psi(L^{1/m}; \beta)y_t + \sum_{k=1}^K \rho_k x_{t,k} + u_t, \quad t \in [T],$$

where $x_{t+1,j}$ is the *Financial crisis* news series. We test whether daily VIX Granger causes future values of the *Financial crisis* news series. Note that we only need to estimate the initial coefficient vector, since the precision matrix remains the same. The results appear in Table 3. They show a rather weak predictability of future news series by daily VIX, suggesting a unidirectional Granger causality pattern.

4.1.3 Granger causal clusters of news topics

The news attention series are classified into 24 broader meta topics that group the individual news series according to a common theme. See again Online Appendix Table OA.1 for details. We test which group of individual news series Granger causes future VIX values. The results are reported in Table 4. They show that the group *Banks* is significant at the 1% significance level. This group consists of news series pertaining to news about

Table 2. VIX Granger causality results

Variable	M_T	M_T^{sG}	M_T^{bt}	M_T^{sG}	M_T^{bt}
		Parzen		QS	
Panel (a)—full sample					
Daily VIX lags		0.000	0.001	0.000	0.003
Financial crisis		0.001	0.001	0.001	0.001
Recession		0.001	0.001	0.002	0.001
Panel (b)—up to 2008					
Daily VIX lags		0.000	0.001	0.000	0.013
Financial crisis		0.008	0.004	0.006	0.002
Recession		0.146	0.117	0.144	0.065
Panel (c)—12 lags					
Daily VIX lags		0.008	0.027	0.013	0.067
Financial crisis		0.011	0.037	0.033	0.023
Recession		0.255	0.434	0.367	0.518

Notes: We report p -values of series that are significant at least at the 5% significance level with two kernel functions, Parzen and QS. Following the rules-of-thumb appearing in Equation (6) we use M_T for sub-Gaussian data, denoted M_T^{sG} , and for heavy-tailed data with $q = 5$, denoted M_T^{bt} .

Table 3. Bi-directional Granger causality results

Variable	M_T	M_T^{sG}	M_T^{bt}	M_T^{sG}	M_T^{bt}
		Parzen		QS	
		Full sample			
Daily VIX lags		0.108	0.105	0.090	0.154

Notes: We report p -values for a range of M_T values and two kernel functions, Parzen and QS. We use M_T for sub-Gaussian data, denoted M_T^{sG} , and for heavy-tailed data with $q = 5$, denoted M_T^{bt} .

Table 4. Group Granger causality results

Variable	M_T	M_T^{sG}	M_T^{bt}	M_T^{sG}	M_T^{bt}
		Parzen		QS	
Banks		0.002	0.005	0.001	0.001

Notes: We report p -values for a range of M_T values and two kernel functions, Parzen and QS. We use M_T for sub-Gaussian data, denoted M_T^{sG} , and for heavy-tailed data with $q = 5$, denoted M_T^{bt} .

Mortgages, Bank loans, Credit ratings, Nonperforming loans, Savings and loans, and the Financial crisis.

5 Conclusion

This article develops valid inferential methods for high-dimensional time-series regressions estimated with the sg-LASSO estimator that encompasses the LASSO and the group LASSO as special cases. We derive the debiased central limit theorem with the explicit bias correction for the sg-LASSO with serially correlated regression errors. Furthermore, we also study HAC estimators of the long-run variance for low-dimensional groups of regression coefficients and characterize how the optimal bandwidth parameter should scale with the sample size, and the temporal dependence, as well as tails of the data. These results lead to the valid t - and Wald tests for the low-dimensional subset of parameters, such as Granger causality tests. Our treatment relies on a new suitable variation of the Fuk-Nagaev inequality for τ -mixing processes which allows us to handle the time-series data with polynomial tails. An interesting avenue for future research is to study more carefully the problem of the optimal data-driven bandwidth choice based on higher-order asymptotic expansions, see, for example, [Sun, Phillips, and Jin \(2008\)](#) for steps in this direction in low-dimensional settings.

In an empirical application, we use a high-dimensional news attention series to study causal patterns between the VIX, sometimes called the fear index, and financial news. We find that almost exclusively the topic of financial crisis exhibits unidirectional Granger causality for the VIX.

Supplemental Data

[Supplemental data](https://www.datahostingsite.com) are available at <https://www.datahostingsite.com>.

Appendix

A.1 Proofs

Proof of Theorem 2.1: By Fermat's rule, the sg-LASSO satisfies

$$\mathbf{X}^\top (\mathbf{X}\hat{\beta} - \mathbf{y})/T + \lambda z^* = 0$$

for some $z^* \in \partial\Omega(\hat{\beta})$, where $\partial\Omega(\hat{\beta})$ is the sub-differential of $b \mapsto \Omega(b)$ at $\hat{\beta}$. Rearranging this expression and multiplying by $\hat{\Theta}$

$$\hat{\beta} - \beta + \hat{\Theta}\lambda z^* = \hat{\Theta}\mathbf{X}^\top \mathbf{u}/T + (I - \hat{\Theta}\hat{\Sigma})(\hat{\beta} - \beta) + \hat{\Theta}\mathbf{X}^\top (\mathbf{m} - \mathbf{X}\beta)/T,$$

where we use $\mathbf{y} = \mathbf{m} + \mathbf{u}$. Plugging in λz^* and multiplying by $\sqrt{T}$

$$\begin{aligned} \sqrt{T}(\hat{\beta} - \beta + B) &= \hat{\Theta}\mathbf{X}^\top \mathbf{u}/\sqrt{T} + \sqrt{T}(I - \hat{\Theta}\hat{\Sigma})(\hat{\beta} - \beta) + \hat{\Theta}\mathbf{X}^\top (\mathbf{m} - \mathbf{X}\beta)/\sqrt{T} \\ &= \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \Theta x_t + \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t (\hat{\Theta} - \Theta) X_t + \sqrt{T}(I - \hat{\Theta}\hat{\Sigma})(\hat{\beta} - \beta) \\ &\quad + \hat{\Theta}\mathbf{X}^\top (\mathbf{m} - \mathbf{X}\beta)/\sqrt{T}. \end{aligned}$$

Next, we look at coefficients corresponding to $G \subset [p]$

$$\begin{aligned}\sqrt{T}(\hat{\beta}_G - \beta_G + B_G) &= \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t \Theta_G x_t + \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t (\hat{\Theta}_G - \Theta_G) x_t + \sqrt{T}(\mathbf{I} - \hat{\Theta} \hat{\Sigma})_G (\hat{\beta} - \beta) \\ &\quad + \hat{\Theta}_G \mathbf{X}^\top (\mathbf{m} - \mathbf{X}\beta) / \sqrt{T} \\ &\triangleq \text{I}_T + \text{II}_T + \text{III}_T + \text{IV}_T.\end{aligned}$$

We will show that $\text{I}_T \xrightarrow{d} N(0, \Xi_G)$ by the triangular array CLT; see [Neumann \(2013\)](#), Theorem 2.1. To that end, by the Crámer-Wold theorem, it is sufficient to show that $\mathbf{z}^\top \text{I}_T \xrightarrow{d} \mathbf{z}^\top N(0, \Xi_G)$ for every $\mathbf{z} \in \mathbf{R}^{|\mathbf{G}|}$. Note that under Assumptions 2.1 and 2.5(i)–(ii)

$$\begin{aligned}\sum_{t=1}^T \mathbb{E} \left| \frac{\mathbf{z}^\top \xi_t}{\sqrt{T}} \right|^2 &= \mathbb{E} |u_t \mathbf{z}^\top \Theta_G x_t|^2 \\ &\leq C \mathbf{z}^\top \Theta_G \Sigma \Theta_G^\top \mathbf{z} \\ &= O(1).\end{aligned}$$

Therefore, since $q > 2r/(r-2)$, we have $\varsigma > 2$, and for every $\epsilon > 0$

$$\sum_{t=1}^T \mathbb{E} \left[\left| \frac{\mathbf{z}^\top \xi_t}{\sqrt{T}} \right|^2 \mathbf{1} \left\{ |\mathbf{z}^\top \xi_t| > \epsilon \sqrt{T} \right\} \right] \leq \frac{\mathbb{E} |\mathbf{z}^\top \xi_t|^\varsigma}{(\epsilon \sqrt{T})^{\varsigma-2}} = o(1).$$

Next, under Assumptions 2.1 and 2.5(i) and (ii), the long-run variance

$$\lim_{T \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T \mathbf{z}^\top \xi_t \right) = \mathbf{z}^\top \Xi_G \mathbf{z}$$

exists by Proposition A.1.1.

Next, put $\mathcal{M} = \sigma(\xi_0, \xi_{-1}, \xi_{-2}, \dots)$, $Y = g(\mathbf{z}^\top \xi_{t_1-t_b}/\sqrt{T}, \dots, \mathbf{z}^\top \xi_0/\sqrt{T}) \mathbf{z}^\top \xi_0$, and $X = \mathbf{z}^\top \xi_r$ for some $t_b \geq 0$. Note that X and $|XY|$ are integrable and that Y is $\mathcal{M}$ -measurable. Therefore, for every measurable function $g: \mathbf{R}^b \rightarrow \mathbf{R}$ with $\sup_x |g(x)| \leq 1$, by [Dedecker and Doukhan \(2003\)](#), Proposition 1, for all $b \in \mathbf{N}$ and all indices $1 \leq t_1 < t_2 < \dots < t_b < t_b + r \leq t_b + s \leq T$

$$\begin{aligned}& \left| \text{Cov} \left(g(\mathbf{z}^\top \xi_{t_1}/\sqrt{T}, \dots, \mathbf{z}^\top \xi_{t_b}/\sqrt{T}) \mathbf{z}^\top \xi_{t_b}/\sqrt{T}, \mathbf{z}^\top \xi_{t_b+r}/\sqrt{T} \right) \right| \\ &= \frac{1}{T} |\text{Cov}(Y, X)| \\ &\leq \frac{1}{T} \int_0^{\gamma(\mathcal{M}, \mathbf{z}^\top \xi_r)} Q_Y \circ G_{\mathbf{z}^\top \xi_r}(u) du \\ &\leq \frac{1}{T} \int_0^{\gamma(\mathcal{M}, \mathbf{z}^\top \xi_r)} Q_{\mathbf{z}^\top \xi_0} \circ G_{\mathbf{z}^\top \xi_r}(u) du \\ &\leq \frac{1}{T} \|\mathbb{E}(\mathbf{z}^\top \xi_r | \mathcal{M}) - \mathbb{E}(\mathbf{z}^\top \xi_r)\|_1^{\varsigma-2} \|\mathbf{z}^\top \xi_0\|_\varsigma^{\varsigma/(\varsigma-1)} \\ &\leq \frac{1}{T} \|\Theta_G^\top \mathbf{z}\|_1^{\varsigma-2} \tau_r^{\varsigma-2} \|\mathbf{z}^\top \xi_0\|_\varsigma^{\varsigma/(\varsigma-1)} \lesssim r^{-a_{\varsigma-2}^{\varsigma-2}}\end{aligned}$$

where the second line follows by stationarity and $\sup_x |g(x)| \leq 1$, the fourth by Hölder's inequality, and the change of variables

$$\int Q_{\mathbf{z}^\top \xi_0}^{-1} \circ G_{\mathbf{z}^\top \xi_r}(u) du = \int_0^1 Q_{\mathbf{z}^\top \xi_0}^\varsigma(u) du = \|\mathbf{z}^\top \xi_0\|_\varsigma^\varsigma,$$

and the last by Lemma A.1.1 and Assumptions 2.1(ii) and 2.5(ii). Similarly,

$$\begin{aligned}
 & \left| \text{Cov} \left(g(z^\top \xi_{t_1} / \sqrt{T}, \dots, z^\top \xi_{t_b} / \sqrt{T}), z^\top \xi_{t_b+r} / \sqrt{T} z^\top \xi_{t_b+s} / \sqrt{T} \right) \right| \\
 &= \frac{1}{T} \left| \text{Cov} \left(g(z^\top \xi_{t_1-t_b} / \sqrt{T}, \dots, z^\top \xi_0 / \sqrt{T}), z^\top \xi_r z^\top \xi_s \right) \right| \\
 &\leq \frac{1}{T} \int_0^{\gamma(\mathcal{M}, z^\top \xi_r z^\top \xi_s)} Q_g \circ G_{z^\top \xi_r z^\top \xi_s}(u) du \\
 &\leq \frac{1}{T} \|\mathbb{E}(z^\top \xi_r z^\top \xi_s | \mathcal{M}) - \mathbb{E}(z^\top \xi_r z^\top \xi_s)\|_1 \\
 &\leq \frac{1}{T} \|\Theta_G^\top z\|_1^2 r \leq r^{-d}.
 \end{aligned}$$

Since the sequence $(r^{-a(\varsigma-2)/(\varsigma-1)\wedge d})_{r \in \mathbb{N}}$ is summable under Assumption 2.1(ii), all conditions of [Neumann \(2013\)](#), Theorem 2.1, are verified, whence $z^\top \mathbf{I}_T \xrightarrow{d} z^\top N(0, \Xi_G)$ for every $z \in \mathbf{R}^{[G]}$.

Next,

$$\begin{aligned}
 |\text{II}_T|_\infty &= \left| (\hat{\Theta} - \Theta)_G \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T u_t x_t \right) \right|_\infty \\
 &\leq \|\hat{\Theta}_G - \Theta_G\|_\infty \left| \frac{1}{\sqrt{T}} \sum_{t=1}^T u_t x_t \right|_\infty \\
 &= O_P \left(\frac{Sp^{1/\kappa}}{T^{1-1/\kappa}} \vee S \sqrt{\frac{\log p}{T}} \right) O_P \left(\frac{p^{1/\kappa}}{T^{1/2-1/\kappa}} + \sqrt{\log p} \right) \\
 &= o_P(1),
 \end{aligned}$$

where the second line follows by $|Ax|_\infty \leq \|A\|_\infty |x|_\infty$, the third line by [Proposition OA.1.2 of the Online Appendix](#) and the inequality in [Equation \(7\)](#) under Assumption 2.1, and the last under Assumption 2.5(iv). Likewise, using $|Ax|_\infty \leq \max_{j,k} |A_{j,k}|_\infty |x|_1$, by [Proposition OA.1.2 and Theorem OA.1 of the Online Appendix](#)

$$\begin{aligned}
 |\text{III}_T|_\infty &= \sqrt{T} |(I - \hat{\Theta} \hat{\Sigma})_G (\hat{\beta} - \beta)|_\infty \\
 &\leq \sqrt{T} \max_{j \in G} |(I - \hat{\Theta} \hat{\Sigma})_j|_\infty |\hat{\beta} - \beta|_1 \\
 &= O_P \left(\frac{p^{1/\kappa}}{T^{1/2-1/\kappa}} \vee \sqrt{\log p} \right) O_P \left(\frac{s_z p^{1/\kappa}}{T^{1-1/\kappa}} \vee s_z \sqrt{\frac{\log p}{T}} \right) \\
 &= o_P(1),
 \end{aligned}$$

under Assumption 2.5(iv). Lastly, by the Cauchy-Schwartz inequality, under Assumption 2.5(v)

$$\begin{aligned}
 |\text{IV}_T|_\infty &\leq \max_{j \in G} |\mathbf{X} \hat{\Theta}_j^\top|_2 \|\mathbf{m} - \mathbf{X} \beta\|_T \\
 &= \max_{j \in G} \sqrt{\hat{\Theta}_j \hat{\Sigma} \hat{\Theta}_j^\top} o_P(1) \\
 &= o_P(1),
 \end{aligned}$$

where the last line follows since $\hat{\Theta}_j$ are consistent for Θ_j in the ℓ_1 norm while $\hat{\Sigma}$ is consistent for Σ in the entrywise maximum norm under the maintained assumptions. $\square$

Next, we focus on the HAC estimator based on LASSO residuals. Note that by construction of the precision matrix $\hat{\Theta}$, its j th row is $\hat{\Theta}_j x_t = \hat{v}_{t,j} / \hat{\sigma}_j^2$, where $\hat{v}_{t,j}$ is the regression residual from the j th nodewise LASSO regression and $\hat{\sigma}_j^2$ is the corresponding estimator of the variance of the regression error. Therefore, the HAC estimator based on the LASSO residuals in [Equation \(3\)](#) can be written as

$$\hat{\Xi}_G = \sum_{|k| < T} K\left(\frac{k}{M_T}\right) \hat{\Gamma}_k,$$

where $\hat{\Gamma}_k$ has generic (j, b) -entry $\frac{1}{T} \sum_{t=1}^{T-k} \hat{u}_t \hat{u}_{t+k} \hat{v}_{t,j} \hat{v}_{t+k,b} \hat{\sigma}_j^{-2} \hat{\sigma}_b^{-2}$.

Similarly, we define

$$\tilde{\Xi}_G = \sum_{|k| < T} K\left(\frac{k}{M_T}\right) \tilde{\Gamma}_k,$$

where $\tilde{\Gamma}_k$ has generic (j, b) -entry $\frac{1}{T} \sum_{t=1}^{T-k} u_t u_{t+k} v_{t,j} v_{t+k,b} \sigma_j^{-2} \sigma_b^{-2}$ and note that the long-run variance Ξ_G has generic (j, b) -entry $\mathbb{E}[u_t u_{t+k} v_{t,j} v_{t+k,b}] \sigma_j^{-2} \sigma_b^{-2}$.

Assumption A.1.1: Suppose that uniformly over $k \in \mathbb{Z}$ and $j, b \in G$ (i) $\mathbb{E}|u_0 u_k v_{0,j} v_{k,b}| < \infty$; (ii) $\mathbb{E}|v_{0,j} u_k v_{k,b}|^2 < \infty$, $\mathbb{E}|u_0 u_k v_{k,b}|^2 < \infty$, $\mathbb{E}|u_0 v_{0,j} u_k|^2 < \infty$, and $\mathbb{E}|u_0 v_{0,j} v_{k,b}|^2 < \infty$; (iii) $\mathbb{E}|u_0|^{2q} < \infty$ and $\mathbb{E}|v_{0,j}|^{2q} < \infty$ for some $q \geq 1$.

Proof of Theorem 2.2: By Proposition OA.1.3 in the Online Appendix with $V_t = (u_t v_{t,j} / \sigma_j^2)_{j \in G}$

$$\|\hat{\Xi}_G - \Xi_G\| \leq \|\hat{\Xi}_G - \tilde{\Xi}_G\| + O_p\left(\sqrt{\frac{M_T}{T}} + M_T^{-\varsigma} + T^{-(\varsigma \wedge 1)}\right). \quad (\text{A.1})$$

Next,

$$\begin{aligned} \|\hat{\Xi}_G - \tilde{\Xi}_G\| &\leq \sum_{|k| < T} \left| K\left(\frac{k}{M_T}\right) \right| \|\hat{\Gamma}_k - \tilde{\Gamma}_k\| \\ &\leq |G| \sum_{|k| < T} \left| K\left(\frac{k}{M_T}\right) \right| \max_{j,b \in G} \left| \frac{1}{\hat{\sigma}_j^2 \hat{\sigma}_b^2 T} \sum_{t=1}^{T-k} \hat{u}_t \hat{u}_{t+k} \hat{v}_{t,j} \hat{v}_{t+k,b} \right. \\ &\quad \left. - \frac{1}{\sigma_j^2 \sigma_b^2 T} \sum_{t=1}^{T-k} u_t u_{t+k} v_{t,j} v_{t+k,b} \right| \\ &\leq |G| \sum_{|k| < T} \left| K\left(\frac{k}{M_T}\right) \right| \max_{j,b \in G} \frac{1}{\hat{\sigma}_j^2 \hat{\sigma}_b^2} \left| \frac{1}{T} \sum_{t=1}^{T-k} \hat{u}_t \hat{u}_{t+k} \hat{v}_{t,j} \hat{v}_{t+k,b} - \frac{1}{T} \sum_{t=1}^{T-k} u_t u_{t+k} v_{t,j} v_{t+k,b} \right| \\ &\quad + |G| \max_{j,b \in G} \left| \frac{1}{\hat{\sigma}_j^2 \hat{\sigma}_b^2} - \frac{1}{\sigma_j^2 \sigma_b^2} \right| \sum_{|k| < T} \left| K\left(\frac{k}{M_T}\right) \right| \left| \frac{1}{T} \sum_{t=1}^{T-k} u_t u_{t+k} v_{t,j} v_{t+k,b} \right| \triangleq S_T^a + S_T^b. \end{aligned}$$

By Proposition OA.1.1 of the Online Appendix, since $s_x^2 \log p / T \rightarrow 0$ and $s_x^2 p / T^{4\kappa/5-1} \rightarrow 0$, under stated assumptions, we obtain $\max_{j \in G} |\hat{\sigma}_j^2 - \sigma_j^2| = o_p(1)$, and whence $\max_{j \in G} \hat{\sigma}_j^{-2} = O_p(1)$. Using $\hat{a}\hat{b} - ab = (\hat{a} - a)b + a(\hat{b} - b) + (\hat{a} - a)(\hat{b} - b)$, by Proposition OA.1.1 of the Online Appendix

$$S_T^b = O_p\left(\frac{s_x p^{1/\kappa}}{T^{1-1/\kappa}} \vee s_x \sqrt{\frac{\log p}{T}}\right) \sum_{|k| < T} \left| K\left(\frac{k}{M_T}\right) \right| \max_{j,b \in G} \left| \frac{1}{T} \sum_{t=1}^{T-k} u_t u_{t+k} v_{t,j} v_{t+k,b} \right|.$$

Under Assumption A.1.1(i) and Assumption OA.1.1(i) in the Online Appendix

$$\begin{aligned} & \mathbb{E} \left[\sum_{|k| < T} \left| K \left(\frac{k}{M_T} \right) \right| \max_{j,b \in G} \left| \frac{1}{T} \sum_{t=1}^{T-k} \mathbf{u}_t \mathbf{u}_{t+k} \mathbf{v}_{t,j} \mathbf{v}_{t+k,b} \right| \right] \\ & \leq O(M_T) \sup_{k \in \mathbb{Z}} \sum_{j,b \in G} \mathbb{E} \left| \frac{1}{T} \sum_{t=1}^{T-k} \mathbf{u}_t \mathbf{u}_{t+k} \mathbf{v}_{t,j} \mathbf{v}_{t+k,b} \right| \\ & \leq O(M_T) |G|^2 \sup_{k \in \mathbb{Z}} \max_{j,b \in G} \mathbb{E} |\mathbf{u}_t \mathbf{u}_{t+k} \mathbf{v}_{t,j} \mathbf{v}_{t+k,b}| \\ & = O(M_T), \end{aligned}$$

and whence $S_T^b = O_P \left(M_T \left(\frac{s_\alpha p^{1/\kappa}}{T^{1-1/\kappa}} \vee s_\alpha \sqrt{\frac{\log p}{T}} \right) \right)$.

Next, we evaluate uniformly over $|k| < T$

$$\begin{aligned} & \left| \frac{1}{T} \sum_{t=1}^{T-k} \hat{\mathbf{u}}_t \hat{\mathbf{u}}_{t+k} \hat{\mathbf{v}}_{t,j} \hat{\mathbf{v}}_{t+k,b} - \frac{1}{T} \sum_{t=1}^{T-k} \mathbf{u}_t \mathbf{u}_{t+k} \mathbf{v}_{t,j} \mathbf{v}_{t+k,b} \right| \\ & \leq \left| \frac{1}{T} \sum_{t=1}^{T-k} (\hat{\mathbf{u}}_t \hat{\mathbf{v}}_{t,j} - \mathbf{u}_t \mathbf{v}_{t,j}) \mathbf{u}_{t+k} \mathbf{v}_{t+k,b} \right| + \left| \frac{1}{T} \sum_{t=1}^{T-k} \mathbf{u}_t \mathbf{v}_{t,j} (\hat{\mathbf{u}}_{t+k} \hat{\mathbf{v}}_{t+k,b} - \mathbf{u}_{t+k} \mathbf{v}_{t+k,b}) \right| \\ & \quad + \left| \frac{1}{T} \sum_{t=1}^{T-k} (\hat{\mathbf{u}}_t \hat{\mathbf{v}}_{t,j} - \mathbf{u}_t \mathbf{v}_{t,j}) (\hat{\mathbf{u}}_{t+k} \hat{\mathbf{v}}_{t+k,b} - \mathbf{u}_{t+k} \mathbf{v}_{t+k,b}) \right| \triangleq \text{I}_T + \text{II}_T + \text{III}_T. \end{aligned}$$

We bound the first term as

$$\begin{aligned} \text{I}_T & \leq \left| \frac{1}{T} \sum_{t=1}^{T-k} (\hat{\mathbf{u}}_t - \mathbf{u}_t) \mathbf{v}_{t,j} \mathbf{u}_{t+k} \mathbf{v}_{t+k,b} \right| + \left| \frac{1}{T} \sum_{t=1}^{T-k} \mathbf{u}_t (\hat{\mathbf{v}}_{t,j} - \mathbf{v}_{t,j}) \mathbf{u}_{t+k} \mathbf{v}_{t+k,b} \right| \\ & \quad + \left| \frac{1}{T} \sum_{t=1}^{T-k} (\hat{\mathbf{u}}_t - \mathbf{u}_t) (\hat{\mathbf{v}}_{t,j} - \mathbf{v}_{t,j}) \mathbf{u}_{t+k} \mathbf{v}_{t+k,b} \right| \triangleq \text{I}_T^a + \text{I}_T^b + \text{I}_T^c. \end{aligned}$$

By the Cauchy–Schwartz inequality, under Assumptions of [Theorem OA.1 in the Online Appendix](#) for the sg-LASSO and Assumption A.1.1(ii)

$$\begin{aligned} \text{I}_T^a & = \left| \frac{1}{T} \sum_{t=1}^{T-k} \left(\mathbf{x}_t^\top (\hat{\beta} - \beta) + m_t - \mathbf{x}_t^\top \beta \right) \mathbf{v}_{t,j} \mathbf{u}_{t+k} \mathbf{v}_{t+k,b} \right| \\ & \leq (\|\mathbf{X}(\hat{\beta} - \beta)\|_T + \|\mathbf{m} - \mathbf{X}\beta\|_T) \sqrt{\frac{1}{T} \sum_{t=1}^{T-k} \mathbf{v}_{t,j}^2 \mathbf{u}_{t+k}^2 \mathbf{v}_{t+k,b}^2} \\ & = O_P \left(\frac{s_\alpha p^{1/\kappa}}{T^{1-1/\kappa}} \vee \sqrt{\frac{s_\alpha \log p}{T}} \right). \end{aligned}$$

Similarly, under Assumptions of Theorem OA.1 for the nodewise LASSO and Assumption A.1.1(ii)

$$\text{I}_T^b \leq \left(\|\mathbf{X}_{-j}(\hat{\gamma}_j - \gamma_j)\|_T + o_P(T^{-1/2}) \right) \sqrt{\frac{1}{T} \sum_{t=1}^{T-k} \mathbf{u}_t^2 \mathbf{u}_{t+k}^2 \mathbf{v}_{t+k,b}^2} = O_P \left(\frac{S_j p^{1/\kappa}}{T^{1-1/\kappa}} \vee \sqrt{\frac{S_j \log p}{T}} \right).$$

Note that for arbitrary $(\xi_t)_{t \in \mathbb{Z}}$ and $q \geq 1$, by Jensen's inequality

$$\mathbb{E}[\max_{t \in [T]} |\xi_t|] \leq (\mathbb{E}[\max_{t \in [T]} |\xi_t|^q])^{1/q} \leq \left(\mathbb{E} \left[\sum_{t=1}^T |\xi_t|^q \right] \right)^{1/q} = T^{1/q} (\mathbb{E}|\xi_t|^q)^{1/q}.$$

Then by the Cauchy–Schwartz inequality under Assumption A.1.1(iii) and [Theorem OA.1 of the Online Appendix](#)

$$\begin{aligned} I_T^c &\leq (||\mathbf{X}(\hat{\beta} - \beta)||_T + o_P(T^{-1/2})) (||\mathbf{X}_{-j}(\hat{\gamma}_j - \gamma_j)||_T + o_P(T^{-1/2})) \max_{t \in [T]} |u_t v_{t,b}| \\ &= O_P\left(\frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s \log p}{T^{1-1/\kappa}}\right), \end{aligned}$$

where we use the fact that $\kappa \leq q$. Therefore, under maintained assumptions

$$I_T = O_P\left(\frac{sp^{1/\kappa}}{T^{1-1/\kappa}} \vee \sqrt{\frac{s \log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s \log p}{T^{1-1/\kappa}}\right)$$

and by symmetry

$$\Pi_T = O_P\left(\frac{sp^{1/\kappa}}{T^{1-1/\kappa}} \vee \sqrt{\frac{s \log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s \log p}{T^{1-1/\kappa}}\right).$$

Lastly, by the Cauchy–Schwartz inequality

$$\begin{aligned} \text{III}_T &\leq \sqrt{\frac{1}{T} \sum_{t=1}^{T-k} (\hat{u}_t \hat{v}_{t,j} - u_t v_{t,j})^2} \sqrt{\frac{1}{T} \sum_{t=1}^{T-k} (\hat{u}_{t+k} \hat{v}_{t+k,b} - u_{t+k} v_{t+k,b})^2} \\ &\leq \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{u}_t \hat{v}_{t,j} - u_t v_{t,j})^2} \sqrt{\frac{1}{T} \sum_{t=1}^T (\hat{u}_t \hat{v}_{t,b} - u_t v_{t,b})^2}. \end{aligned}$$

For each $j \in G$

$$\begin{aligned} \frac{1}{T} \sum_{t=1}^T (\hat{u}_t \hat{v}_{t,j} - u_t v_{t,j})^2 &\leq \frac{3}{T} \sum_{t=1}^T |\hat{u}_t - u_t|^2 v_{t,j}^2 + \frac{3}{T} \sum_{t=1}^T |\hat{v}_{t,j} - v_{t,j}|^2 u_t^2 \\ &\quad + \frac{3}{T} \sum_{t=1}^T |\hat{u}_t - u_t|^2 |\hat{v}_{t,j} - v_{t,j}|^2 \\ &\triangleq \text{III}_T^a + \text{III}_T^b + \text{III}_T^c. \end{aligned}$$

Since under Assumption A.1.1(iii), $\mathbb{E}|v_{t,j}|^{2q} < \infty$ and $\mathbb{E}|u_t|^{2q} < \infty$,

$$\text{III}_T^a \leq 3 \max_{t \in [T]} |v_{t,j}|^2 (||\mathbf{X}(\hat{\beta} - \beta)||_T^2 + o_P(T^{-1/2})) = O_P\left(\frac{s_x p^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s_x \log p}{T^{1-1/\kappa}}\right)$$

and

$$\text{III}_T^b \leq 3 \max_{t \in [T]} |u_t|^2 (||\mathbf{X}_{-j}(\hat{\gamma}_j - \gamma_j)||_T^2 + o_P(T^{-1/2})) = O_P\left(\frac{S_j p^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{S_j \log p}{T^{1-1/\kappa}}\right).$$

For the last term, since under Assumption 2.1(ii), $\sup_k \mathbb{E}|X_{t,k}|^{2\tilde{q}} < \infty$ and $\kappa \geq \tilde{q}$, by [Theorem OA.1 of the Online Appendix](#)

$$\begin{aligned}
 \text{III}_T^c &\leq 3(\|\mathbf{X}(\hat{\beta} - \beta)\|_T^2 + o_P(T^{-1/2})) \max_{t \in [T]} |X_{t,-j}^\top (\hat{\gamma}_j - \gamma_j) + \mathbf{m}_t - X_t^\top \beta|^2 \\
 &\leq O_P\left(\frac{s_x p^{2/\kappa}}{T^{2-2/\kappa}} \vee \frac{s_x \log p}{T}\right) \left(2 \max_{t \in [T]} |X_{t,-j}|^2 |\hat{\gamma}_j - \gamma_j|_1^2 + 2T \|\mathbf{m} - \mathbf{X}^\top \beta\|_T^2\right) \\
 &= O_P\left(\left(\frac{s_x p^{2/\kappa}}{T^{2-2/\kappa}} \vee \frac{s_x \log p}{T}\right) \left(\frac{S^2 p^{2/\kappa}}{T^{2-2/\kappa}} \vee S^2 \frac{\log p}{T}\right) (pT)^{1/\kappa}\right) \\
 &= O_P\left(\frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} + \frac{s^3 p^{3/\kappa} \log p}{T^{3-3/\kappa}} + \frac{s^3 p^{1/\kappa} \log^2 p}{T^{2-1/\kappa}}\right) \\
 &= O_P\left(\frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} + \frac{s^3 p^{3/\kappa} \log p}{T^{3-3/\kappa}}\right),
 \end{aligned}$$

where we use the fact that $\kappa > 2$, $s = s_x \vee S$, $s^\kappa p = o(T^{4\kappa/5-1})$, and $s^2 \log p/T \rightarrow 0$ as $T \rightarrow \infty$. Then for every $j \in G$

$$\frac{1}{T} \sum_{t=1}^T (\hat{u}_t \hat{v}_{t,j} - u_t v_{t,j})^2 = O_P\left(\frac{sp^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s \log p}{T^{1-1/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} + \frac{s^3 p^{3/\kappa} \log p}{T^{3-3/\kappa}}\right),$$

and whence

$$\text{III}_T = O_P\left(\frac{sp^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s \log p}{T^{1-1/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} + \frac{s^3 p^{3/\kappa} \log p}{T^{3-3/\kappa}}\right).$$

Therefore, since $\hat{\sigma}_j^2 \xrightarrow{P} \sigma_j^2$, we obtain

$$\begin{aligned}
 S_T^a &= O_P\left(M_T \left(\frac{sp^{1/\kappa}}{T^{1-1/\kappa}} \vee \sqrt{\frac{s \log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} \vee \frac{s \log p}{T^{1-1/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}} + \frac{s^3 p^{3/\kappa} \log p}{T^{3-3/\kappa}}\right)\right) \\
 &= O_P\left(M_T \left(\frac{sp^{1/\kappa}}{T^{1-1/\kappa}} \vee \sqrt{\frac{s \log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}}\right)\right),
 \end{aligned}$$

where the last line follows since $s^\kappa p/T^{4\kappa/5-1} = o(1)$. Combining this estimate with the previously obtained estimate for S_T^b

$$\|\hat{\Xi}_G - \tilde{\Xi}_G\| = O_P\left(M_T \left(\frac{sp^{1/\kappa}}{T^{1-1/\kappa}} \vee s \sqrt{\frac{\log p}{T}} + \frac{s^2 p^{2/\kappa}}{T^{2-3/\kappa}} + \frac{s^3 p^{5/\kappa}}{T^{4-5/\kappa}}\right)\right).$$

The result follows from combining this estimate with the estimate in Equation (A.1). $\square$

Proof of Theorem 3.1: Suppose first that $p = 1$. For $a \in \mathbf{R}$, with some abuse of notation, let $[a]$ denote its integer part. We split partial sums into blocks $V_k = \xi_{(k-1)J+1} + \dots + \xi_{kJ}$, $k = 1, 2, \dots, [T/J]$ and $V_{[T/J]+1} = \xi_{[T/J]J+1} + \dots + \xi_T$, where we set $V_{[T/J]+1} = 0$ if T/J is an integer. Let $\{U_t : t = 1, 2, \dots, [T/J] + 1\}$ be i.i.d. random variables drawn from the uniform distribution on $(0, 1)$ independently of $\{V_t : t = 1, 2, \dots, [T/J] + 1\}$. Put $\mathcal{M}_t = \sigma(V_1, \dots, V_{t-2})$ for every $t = 3, \dots, [T/J] + 1$. Next, for $t = 1, 2$, set $V_t^* = V_t$, while for $t \geq 3$, by [Dedecker and Prieur \(2004\)](#), Lemma 5, there exist random variables $V_t^* =_d V_t$ such that:

1. V_t^* is $\sigma(V_1, \dots, V_{t-2}) \vee \sigma(V_t) \vee \sigma(U_t)$ -measurable;
2. $V_t^* \perp\!\!\!\perp (V_1, \dots, V_{t-2})$;
3. $\|V_t - V_t^*\|_1 = \tau(\mathcal{M}_t, V_t)$.

It follows from Properties 1 and 2 that $(V_{2t}^*)_{t \geq 1}$ and $(V_{2t-1}^*)_{t \geq 1}$ are sequences of independent random variables. Then

$$\left| \sum_{t=1}^T \xi_t \right| \leq \left| \sum_{t \geq 1} V_{2t}^* \right| + \left| \sum_{t \geq 1} V_{2t-1}^* \right| + \sum_{t=3}^{[T/J]+1} |V_t - V_t^*|$$

$$\triangleq \text{I}_T + \text{II}_T + \text{III}_T.$$

By [Fuk and Nagaev \(1971\)](#), Corollary 4, there exist constants $c_q^{(j)}$, $j = 1, 2$ such that

$$\begin{aligned} \Pr(\text{I}_T \geq x) &\leq \frac{c_q^{(1)}}{x^q} \sum_{t \geq 1} \mathbb{E}|V_{2t}^*|^q + 2 \exp \left(-\frac{c_q^{(2)} x^2}{\sum_{t \geq 1} \text{Var}(V_{2t}^*)} \right) \\ &\leq \frac{c_q^{(1)}}{x^q} \sum_{t \geq 1} \mathbb{E}|V_{2t}|^q + 2 \exp \left(-\frac{c_q^{(2)} x^2}{B_T^2} \right), \end{aligned}$$

where the second inequality follows since $\sum_{t \geq 1} \text{Var}(V_{2t}^*) = \sum_{t \geq 1} \text{Var}(V_{2t}) \leq B_T^2$. Similarly

$$\Pr(\text{II}_T \geq x) \leq \frac{c_q^{(1)}}{x^q} \sum_{t \geq 1} \mathbb{E}|V_{2t-1}|^q + 2 \exp \left(-\frac{c_q^{(2)} x^2}{B_T^2} \right).$$

Lastly, by Markov's inequality and property 3

$$\begin{aligned} \Pr(\text{III}_T \geq x) &\leq \frac{1}{x} \sum_{t=3}^{[T/J]+1} \tau(\mathcal{M}_t, V_t) \\ &\leq \frac{1}{x} \sum_{t=3}^{[T/J]+1} \tau(\mathcal{M}_t, (\xi_{(t-1)J+1}, \dots, \xi_{tJ})) \\ &\leq \frac{1}{x} [T/J] \sup_{t+J+1 \leq t_1 < \dots < t_J} \tau(\mathcal{M}_t, (\xi_{t_1}, \dots, \xi_{t_J})) \\ &\leq \frac{T}{x} \tau_{J+1}, \end{aligned}$$

where the second inequality follows since the sum is a 1-Lipschitz function with respect to $|\cdot|_1$ -norm and the third since $\mathcal{M}_t$ and $(\xi_{(t-1)J+1}, \dots, \xi_{tJ})$ are separated by $J+1$ lags of $(\xi_t)_{t \in \mathbb{Z}}$.

Combining all the estimates together

$$\begin{aligned} \Pr \left(\left| \sum_{t=1}^T \xi_t \right| \geq 3x \right) &\leq \Pr(\text{I}_T \geq x) + \Pr(\text{II}_T \geq x) + \Pr(\text{III}_T \geq x) \\ &\leq \frac{c_q^{(1)}}{x^q} \sum_{t=1}^{[T/J]+1} \mathbb{E}|V_t|^q + 4 \exp \left(-\frac{c_q^{(2)} x^2}{B_T^2} \right) + \frac{T}{x} \tau_{J+1} \\ &\leq \frac{c_q^{(1)}}{x^q} J^{q-1} \sum_{t=1}^T \|\xi_t\|_q^q + \frac{T}{x} c(J+1)^{-a} + 4 \exp \left(-\frac{c_q^{(2)} x^2}{B_T^2} \right). \end{aligned}$$

To balance the first two terms, we shall set $J \sim x^{\frac{q-1}{q+a-1}}$, in which case we obtain the result under maintained assumptions. The result for $p > 1$ follows by the union bound. $\square$

For a stationary process $(\xi_t)_{t \in \mathbb{Z}}$, let

$$\gamma_k = \|\mathbb{E}(\xi_k | \mathcal{M}_0) - \mathbb{E}(\xi_k)\|_1$$

be its L_1 mixingale coefficient with respect to the canonical filtration $\mathcal{M}_0 = \sigma(\xi_0, \xi_{-1}, \xi_{-2}, \dots)$. Let α_k be the α -mixing coefficient and let Q be the quantile function of $|\xi_0|$. The following covariance inequality allows us controlling the autocovariances

in terms of the τ -mixing coefficient as well as comparing the latter to the mixingale and the α -mixing coefficients.

Lemma A.1.1: *Let $(\xi_t)_{t \in \mathbb{Z}}$ be a centered stationary stochastic process with $\|\xi_0\|_q < \infty$ for some $q > 2$. Then*

$$|\text{Cov}(\xi_0, \xi_t)| \leq \gamma_t^{\frac{q-2}{q-1}} \|\xi_0\|_q^{q/(q-1)}$$

and

$$\gamma_t \leq \tau_t \leq 2 \int_0^{2\tau_t} Q(u) du.$$

Proof: Let G be the generalized inverse of $x \mapsto \int_0^x Q(u) du$. By [Dedecker and Doukhan \(2003\)](#), Proposition 1

$$\begin{aligned} |\text{Cov}(\xi_0, \xi_t)| &\leq \int_0^{\gamma_t} (Q \circ G)(u) du \\ &\leq \gamma_t^{\frac{q-2}{q-1}} \left(\int_0^{\|\xi_0\|_1} (Q \circ G)^{q-1}(u) du \right)^{1/(q-1)} \\ &= \gamma_t^{\frac{q-2}{q-1}} \|\xi_0\|_q^{q/(q-1)}, \end{aligned}$$

where the second line follows by Hölder's inequality and the last equality by the change of variables $\int_0^{\|\xi_0\|_1} (Q \circ G)^{q-1}(u) du = \int_0^1 Q^q(u) du = \mathbb{E}|\xi_0|^q$. The second statement follows from [Dedecker and Doukhan \(2003\)](#), Lemma 1 and [Dedecker and Prieur \(2004\)](#), Lemma 6. $\square$

The following result shows that the variance of partial sums can be controlled provided that the τ -mixing coefficients decline sufficiently fast.

Lemma A.1.2: *Let $(\xi_t)_{t \in \mathbb{Z}}$ be a centered stationary stochastic process such that $\|\xi_t\|_q < \infty$ for some $q > 2$ and $\tau_k = O(k^{-a})$ for some $a > \frac{q-1}{q-2}$. Then*

$$\sum_{t=1}^T \sum_{k=1}^T |\text{Cov}(\xi_{t,j}, \xi_{k,j})| = O(T).$$

Proof: Under stationarity

$$\begin{aligned} \sum_{t=1}^T \sum_{k=1}^T |\text{Cov}(\xi_{t,j}, \xi_{k,j})| &= T \text{Var}(\xi_0) + 2 \sum_{k=1}^{T-1} (T-k) \text{Cov}(\xi_0, \xi_k) \\ &\leq T \text{Var}(\xi_0) + 2T \|\xi_0\|_q^{q/(q-1)} \sum_{k=1}^{T-1} \tau_k^{\frac{q-2}{q-1}} \\ &= O(T), \end{aligned}$$

where the second line follows by Proposition A.1.1 and the last since the series $\sum_{k=1}^{\infty} k^{-a \frac{q-2}{q-1}}$ converges under the maintained assumptions. $\square$

Lastly, we show that in the linear regression setting, the long-run variance for a group of projection coefficients $G \subset [p]$ of fixed size exists under mild conditions. Let

$\xi_t = u_t \Theta_G x_t$, where Θ_G are rows of the precision matrix $\Theta = \Sigma^{-1}$ corresponding to indices in G .

Proposition A.1.1: Suppose that (i) $(u_t x_t)_{t \in \mathbb{Z}}$ is stationary for every $p \geq 1$; (ii) $\|u_t\|_q < \infty$ and $\max_{j \in [p]} \|x_{t,j}\|_r = O(1)$ for some $q > 2r/(r-2)$ and $r > 4$; (iii) for every $j \in [p]$, the τ -mixing coefficients of $(u_t x_{t,j})_{t \in \mathbb{Z}}$ are $\tau_k \leq ck^{-a}$ for all $k \geq 0$, where $c > 0$ and $a > (\varsigma - 1)/(\varsigma - 2)$, and $\varsigma = qr/(q+r)$ are some universal constants; (iv) $\|\Theta_G\|_\infty = O(1)$ and $\sup_x \mathbb{E}[|u_t|^2 | x_t = x] = O(1)$. Then for every $z \in \mathbf{R}^{|G|}$, the limit

$$\lim_{T \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T z^\top \xi_t \right)$$

exists.

Proof: Under Assumption (i), by Hölder's inequality, $\max_{j \in [p]} \|u_t x_{t,j}\|_\varsigma = O(1)$ with $\varsigma = qr/(q+r)$, whence by the Minkowski inequality and Assumption (iv)

$$\|z^\top \xi_t\|_\varsigma \leq \sum_{k \in G} \sum_{j \in [p]} |\Theta_{k,j}| \|u_t x_{t,j}\|_\varsigma \leq |G| \|\Theta_G\|_\infty = O(1). \quad (\text{A.2})$$

Since $\varsigma > 2$, this shows that $\text{Var}(z^\top \xi_0)$ exists. Moreover,

$$\begin{aligned} \text{Var}(z^\top \xi_0) &= z^\top \Theta_G \text{Var}(u_0 x_0) \Theta_G^\top z \\ &= \sum_{j,k \in [p]} (z^\top \Theta_G)_j (z^\top \Theta_G)_k \mathbb{E}[u_0^2 x_{0,j} x_{0,k}], \end{aligned}$$

where the sum converges as $p \rightarrow \infty$ by the comparison test under Assumption (iv) implying that $\lim_{T \rightarrow \infty} \text{Var}(z^\top \xi_0)$ exists. Next, under Assumption (i), for every $z \in \mathbf{R}^{|G|}$

$$\text{Var} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T z^\top \xi_t \right) = \text{Var}(z^\top \xi_0) + 2 \sum_{k=1}^{T-1} \left(1 - \frac{k}{T} \right) \text{Cov}(z^\top \xi_0, z^\top \xi_k).$$

By Lemma A.1.1, we bound covariances by the mixingale coefficient for every $k \geq 1$

$$\begin{aligned} |\text{Cov}(z^\top \xi_0, z^\top \xi_k)| &\leq \|z^\top \xi_0\|_\varsigma^{(\varsigma-1)} \|\mathbb{E}(z^\top \xi_k | \mathcal{M}_0) - \mathbb{E}(z^\top \xi_k)\|_1^{\frac{\varsigma-2}{\varsigma-1}} \\ &\leq \|z^\top \Theta_G\|_1^{\frac{\varsigma-2}{\varsigma-1}} \max_{j \in [p]} \|\mathbb{E}(u_k x_{k,j} | \mathcal{M}_0) - \mathbb{E}(u_k x_{k,j})\|_1^{\frac{\varsigma-2}{\varsigma-1}}, \\ &\leq \tau_k^{\frac{\varsigma-2}{\varsigma-1}} \end{aligned}$$

where the first inequality follows by Lemma A.1.1, the second by Equation (A.2), and the third by Lemma A.1.1. Under Assumption (iii), $\sum_{k=1}^{\infty} \tau_k^{(\varsigma-2)/(\varsigma-1)}$ converges, which implies that

$$\sum_{k=1}^{\infty} |\text{Cov}(z^\top \xi_0, z^\top \xi_k)| < \infty$$

by the comparison test. Therefore, by Lebesgue's dominated convergence, this shows that the long-run variance

$$\lim_{T \rightarrow \infty} \text{Var} \left(\frac{1}{\sqrt{T}} \sum_{t=1}^T z^\top \xi_t \right)$$

exists. □

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